

Mechanics 1

R. TEKERCIOGLU

A = 80% 7.10

20.09.16

2 Midterm + 1 final + 3 Quiz (hobersiz)

↳ %10

Solli
B = 40% 7.10
? Homework = 7.10

$$VizeNotu = \frac{Vize1 + Vize2}{2} + A + B$$

$$VizeNotu \times 60 + Final \times 40 = DSnem Sonu.$$

Contents

- General Principles
- Force Vectors, Force Systems Resultants
- Equilibrium of Rigid Body
- Structural Analysis
 - Center of Gravity and Centroid
 - Moment of Inertia, Parallel Axis Theorem
 - Virtual work
 - Potential Energy and Equilibrium, Stability of Equilibrium

Sources

- R.C Hibbeler, Fan Sou Cheong, "Statics & Mechanics of Materials SI", Prentice Hall, 2011
- Bedford, W.Fowler, Y.Ahmad, "Engineering Mechanics: Statics", Prentice Hall, 2008
- Peter Ogorodnik, "Fundamental Engineering Mechanics", Prentice Hall, 1997

22.09.16

Perzembe

Mechanics

This subject is divided into three branches: - rigid-body mechanics
- deformable body " "

(Elasticity, Strength of materials)

- fluid mechanics

(compressible fluid mech, incompressible fluid mech)

Rigid body mechanics is divided into two areas: - statics

- Dynamics

(Kinetics)

Statics deals with the equilibrium of bodies, that is, those that either at rest or move with a constant velocity; whereas dynamics is concerned with the accelerated motion of bodies.

Basic Quantities:

Length: Length is needed to locate the position of a point in space.

②

Time: Succession of events. (olayların geyekiesme siredi) (T)

Mass: Mass is a property of matter by which we can compare the action of one body with that of another (M)

Force: Force is considered as a "push" or "pull" exerted by one body on another. (cismin digen durumunu deghirtmeye yarayan etki)

System of Units:

Name	Length	Time	Mass	Force
International System of Units (SI)	meter (m)	second (s)	kilogram (kg)	Newton (N) = $\frac{kg \cdot m}{s^2}$
MKS	m	s	kg	$\frac{kg \cdot m}{s^2}$
CGS	cm	s	gr	Din: $\frac{gr \cdot cm}{s^2}$
English	ft	s	lb	Pound: $\frac{lb \cdot ft}{s^2}$

$1N \approx 9.81 kg$; $1in = 2.54 cm$; $1ft = 12in = 30.48 cm$; $1Pa = 1N/m^2$

$1lb = 453.59 gm$; $1kg = 2.2046 lb$; $1Psi = 0.0703 kg/cm^2 = 6.894 kPa$
 $= 6.894 N/m^2$

$Psi = lb/in^2$; Pa = Pascal.

Idealizations: Models or idealizations are used in mechanics in order to simplify application of the theory.

Kislen'vor hoami linal edileler.

Particle: A particle has a mass, but a size that can be neglected.

Rigid Body: A rigid body can be considered as a combination of a large number of particles in which all the particles remain at a fixed distance from one another both before and after applying a load.

Concentrated Force: A concentrated force represents the effect of a loading which is assumed to act at a point on a body. (lastilicörnezi, tek bir noktadan deghiyor ama modellerken kuvveti tek noktadan duymacepiz.)

- Continuous cismilerden bairsetti. totali malumeler; arasinda koqluk alabilir. dolu plai duymaceghit.

Dersler
09:32 da

!!!

09:15 !

seccil deghin
you.
ciz deghil
Annoteti

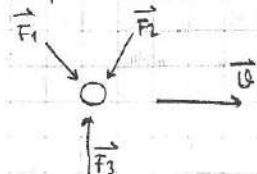
③

Newton's Three Laws of motion:

bir parçacık o zaman
kuvvet etki etmediği
doğru hareket eder

First Law: A particle originally at rest, or moving in a straight line with constant velocity, will remain in this state provided the particle is not subjected to an unbalanced force.

kuvvetlerin
bileşkesi
unbalanced
olmadıkça
durağan kalır



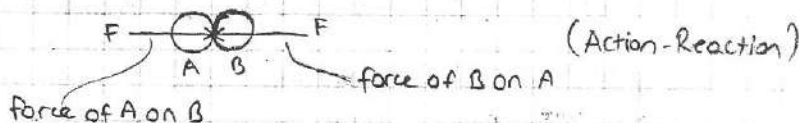
equilibrium

Second Law: A particle acted upon by an unbalanced force $\vec{F}$ experiences an acceleration $\vec{a}$ that has the same direction as the force and a magnitude that is directly proportional to the force $\vec{F}$.

$$\vec{F} = m \cdot \vec{a}$$

accelerated motion

Third Law: The mutual forces of action and reaction between two particles are equal, opposite, and collinear.



Newton's Law of Gravitational Attraction:

$$F = G \frac{m_1 m_2}{r^2}, \text{ where } F = \text{force of gravitation between two particles}$$

G = universal constant of gravitation, according to experimental evidence,

$$G = 66.73 (10^{-12}) \text{ m}^3/\text{kg s}^2$$

m_1, m_2 = mass of each of the particles

r = distance between two particles.

$$W = G \frac{m M_E}{r^2}, \text{ letting } g = G M_E / r^2 \text{ yields } W = mg \Rightarrow F = m \cdot a$$

weight.

④

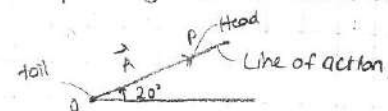
Force Vectors

Scalars and vectors:

Scalar: A quantity characterized by a positive or negative number is called a scalar.

Vector: A vector is a quantity that has both a magnitude and a direction.

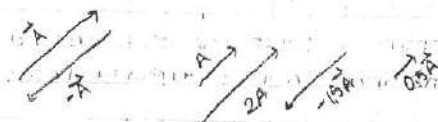
$$\vec{A} \rightarrow |\vec{A}|$$



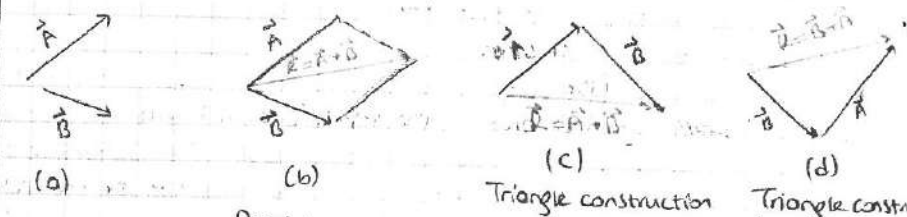
Eğer uyarı, bir
tüm yönü, bir
ve demektir.
Kuvvet vektör: yönlü
değiştirilebilir
ve değişir.

Vector operations:

Multiplication and Division of a vector by a scalar:

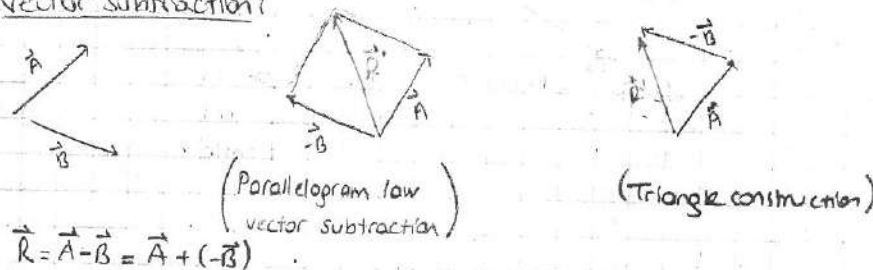


Vector addition:



$\vec{R}$: resultant vector. (bileşke vektör)

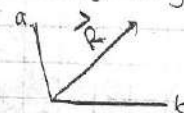
Vector subtraction:



$$\vec{R} = \vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

Resolution of a Vector: (vektör bileşimlerine ayırma)

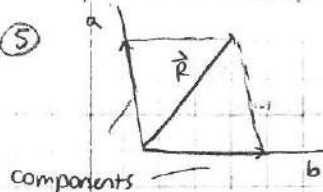
A vector may be resolved into two "components" having known lines of action by using parallelogram law.



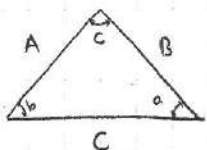
Extend parallel lines from the head of R to form components.

27.09.16 Salı

⑤



Trigonometry:



Sine Law:

$$\frac{A}{\sin \alpha} = \frac{B}{\sin \beta} = \frac{C}{\sin \gamma}$$

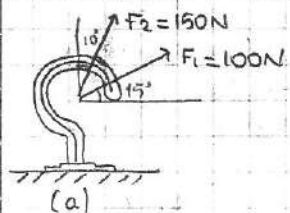
cos Law:

$$C = \sqrt{A^2 + B^2 - 2AB \cos \gamma}$$

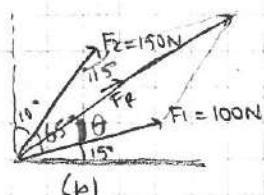
* The magnitude of the resultant force can be determined from the law of cosine, and the direction is determined from the law of sine.

* The magnitude of two force components are determined from the law of Sine.

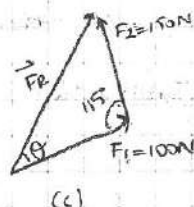
EX:



The screw eye in Fig. is subjected to two forces, F_1 and F_2 . Determine the magnitude and direction of the resultant force.



parallelogram law



Triangle construction.

$\gamma = 115^\circ$

By using cosine law:

$$F_R = \sqrt{F_1^2 + F_2^2 - 2F_1F_2 \cos 115^\circ} = \sqrt{100^2 + 150^2 - 2(100)(150)(-0.4226)} = 212.6 \text{ N}$$

By using sine law:

$$\frac{F_R}{\sin 115^\circ} = \frac{F_2}{\sin \theta} = \frac{F_1}{\sin \gamma} \Rightarrow \frac{212.6}{\sin 115^\circ} = \frac{150}{\sin \theta} \Rightarrow \sin \theta = \frac{(150 \sin 115^\circ)}{212.6}$$

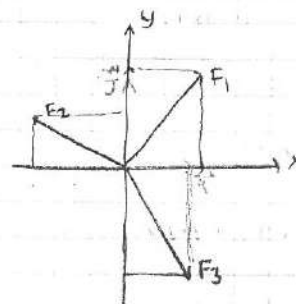
⑥

$$\theta = \sin^{-1} \left(\frac{150 (\sin 115^\circ)}{212.6} \right) = 39.8^\circ$$

Thus, the direction ϕ of F_R , measured from the horizontal, is

$$\phi = 39.8^\circ + 15^\circ = 54.8^\circ$$

Addition of a system of coplanar Forces:



if $\vec{F}_1 = F_{1x}\vec{i} + F_{1y}\vec{j}$

$\vec{F}_2 = -F_{2x}\vec{i} + F_{2y}\vec{j}$

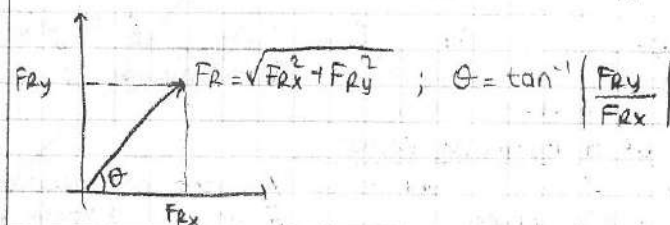
$\vec{F}_3 = F_{3x}\vec{i} - F_{3y}\vec{j}$

the vector resultant is

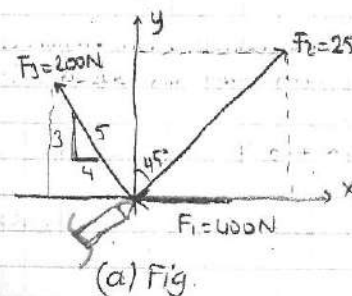
$$\vec{F}_R = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 \Rightarrow \vec{F}_R = (F_{1x}\vec{i} + F_{1y}\vec{j}) + (-F_{2x}\vec{i} + F_{2y}\vec{j}) + (F_{3x}\vec{i} - F_{3y}\vec{j})$$

In general case $F_{Rx} = \sum F_x$, $F_{Ry} = \sum F_y$

$$\vec{F}_R = (F_{1x} - F_{2x} + F_{3x})\vec{i} + (F_{1y} + F_{2y} - F_{3y})\vec{j}$$



The end of the truss in Fig. is subjected to three construct and coplanar forces. Determine the magnitude and orientation of the resultant force.

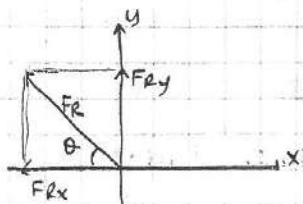


$$\begin{aligned} \rightarrow F_{Rx} = \sum F_x \Rightarrow F_{Rx} &= -400 + F_2 \cos 45^\circ - F_3 \frac{4}{5} \\ &= -400 \text{ N} + (250 \text{ N}) \frac{1}{\sqrt{2}} - (200 \text{ N}) \frac{4}{5} = -383.2 \text{ N} \\ \uparrow F_{Ry} = \sum F_y \Rightarrow F_{Ry} &= F_2 \sin 45^\circ + F_3 \frac{3}{5} \\ &= (250 \text{ N}) \frac{1}{\sqrt{2}} + (200 \text{ N}) \frac{3}{5} = 296.8 \text{ N} \end{aligned}$$

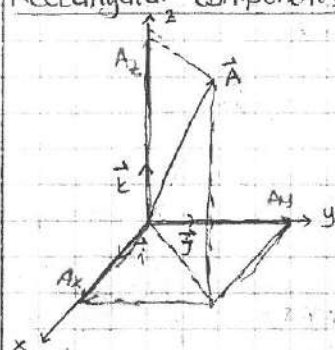
⑦

The resultant force

$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} = \sqrt{(-383.2)^2 + (296.8)^2} = 485 \text{ N}$$

The orientation of F_R :

$$\theta = \tan^{-1} \left(\left| \frac{F_{Ry}}{F_{Rx}} \right| \right) = \tan^{-1} \left(\frac{296.8}{383.2} \right) = 37.8^\circ$$

Cartesian VectorsRectangular components of a vector:

$$\vec{A} = A_x \vec{i} + A_y \vec{j} + A_z \vec{k}$$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

(a magnitude of vector $\vec{A}$.)

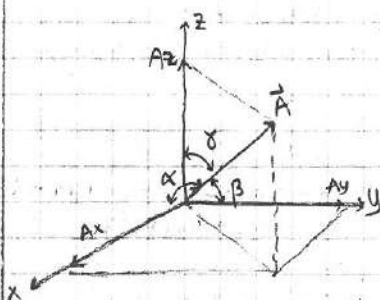
$$\vec{u}_A = \frac{\vec{A}}{|\vec{A}|} = \frac{\vec{A}}{\sqrt{A_x^2 + A_y^2 + A_z^2}}$$

(unit vector form of $\vec{A}$.)

$$= \frac{A_x}{\sqrt{A_x^2 + A_y^2 + A_z^2}} \vec{i} + \frac{A_y}{\sqrt{A_x^2 + A_y^2 + A_z^2}} \vec{j} + \frac{A_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}} \vec{k}$$

Direction of a Cartesian Vector:

The orientation of $\vec{A}$ is defined by the coordinate direction angles α, β, γ measured between the tail of $\vec{A}$ and the positive x, y, z axes located at the tail of $\vec{A}$. ($\alpha, \beta, \gamma < 180^\circ$)



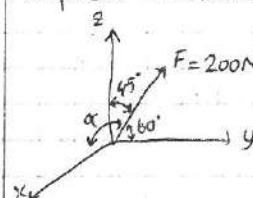
$$\cos \alpha = \frac{A_x}{|\vec{A}|}, \cos \beta = \frac{A_y}{|\vec{A}|}, \cos \gamma = \frac{A_z}{|\vec{A}|}$$

These numbers are known as the direction cosines of $\vec{A}$.

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1 \Rightarrow$$

$$\vec{u}_A = \cos \alpha \vec{i} + \cos \beta \vec{j} + \cos \gamma \vec{k}$$

✓ Ex ⑧

Express the force F shown in Fig. as a cartesian vector.

$$\cos \alpha = \pm 0.5$$

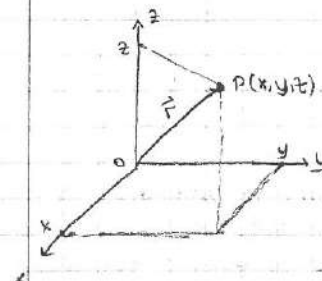
$$C_{\alpha}^2 + C_{\beta}^2 + C_{\gamma}^2 = 1$$

$$\vec{F} = \left(\frac{200}{\cos 45^\circ} \vec{i} + \frac{200}{\cos 60^\circ} \vec{j} + 141.4 \vec{k} \right) \text{ N}$$

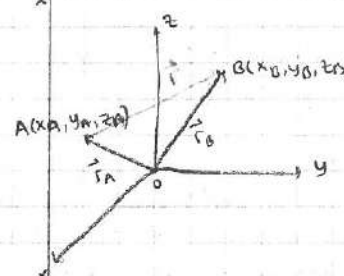
Position Vector:

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The position vector $\vec{r}$ is defined as a fixed vector which locates a point in space relative to another point.



$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$



$$\vec{r}_A + \vec{r} = \vec{r}_B$$

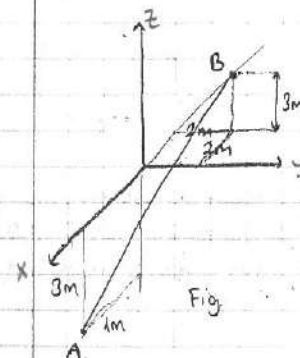
$$\vec{r} = \vec{r}_B - \vec{r}_A$$

$$\vec{r} = (x_B - x_A)\vec{i} + (y_B - y_A)\vec{j} + (z_B - z_A)\vec{k}$$

(position vector from A to B.)

✓ Ex:

An elastic rubber band is attached to points A and B as shown in Fig. Determine its length and its direction measured from A to B.



$$A(1\text{m}, 0, -3\text{m})$$

$$B(-2\text{m}, 2\text{m}, 3\text{m})$$

$$\vec{r}_{AB} = \vec{r}_B - \vec{r}_A = (-2\vec{i} + 2\vec{j} + 3\vec{k}) - (1\vec{i} + 0\vec{j} - 3\vec{k})$$

$$= -3\vec{i} + 2\vec{j} + 6\vec{k}$$

$$|\vec{r}_{AB}| = \sqrt{(-3)^2 + (2)^2 + (6)^2} = 7$$

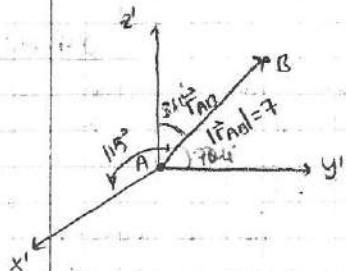
$$\vec{u}_{AB} = \frac{\vec{r}_{AB}}{|\vec{r}_{AB}|} = \frac{-3\vec{i} + 2\vec{j} + 6\vec{k}}{7}$$

9

$$\cos \alpha = -\frac{3}{7} \Rightarrow \alpha = \text{Arccos}\left(-\frac{3}{7}\right) = \cos^{-1}\left(-\frac{3}{7}\right) \Rightarrow \alpha = 115^\circ$$

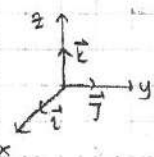
$$\cos \beta = \frac{2}{7} \Rightarrow \beta = \text{Arccos}\left(\frac{2}{7}\right) \Rightarrow \beta = 73.4^\circ$$

$$\cos \delta = \frac{6}{7} \Rightarrow \delta = \text{Arccos}\left(\frac{6}{7}\right) \Rightarrow \delta = 31.0^\circ$$



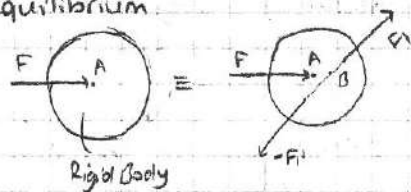
Dot product

The dot product of vectors $\vec{A}$ and $\vec{B}$ is defined as the product of the magnitudes of $\vec{A}$ and $\vec{B}$ and the cosine of the angle θ between their tails.



$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta \text{ where } 0^\circ \leq \theta \leq 180^\circ$$

Law of superposition: The action of a given system of forces on a rigid body will in no way be changed if we add to or subtract from them another system of forces in equilibrium.



(Equilibrium)

Remark: If a body known to be in equilibrium is acted upon by several concurrent, coplanar forces, then these forces, or rather their free vectors, when geometrically must form a closed polygon. (kuvvetler kapalı)

engde
olun boyca
kuvvetin
etkine
ya da bir
maddi
cisim
kuvveti
kuvveti.

10

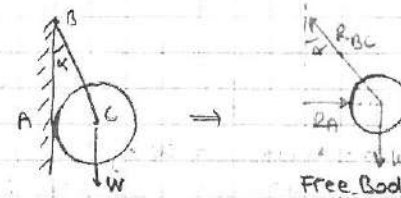
Equilibrium of a particle

A particle is in equilibrium provided it is at rest if originally at rest or has constant velocity if originally in motion. To maintain equilibrium, it is necessary to satisfy Newton's first law of motion, which requires the resultant force acting on a particle to be equal to zero.

$$\sum \vec{F} = 0$$

Free-Body Diagram

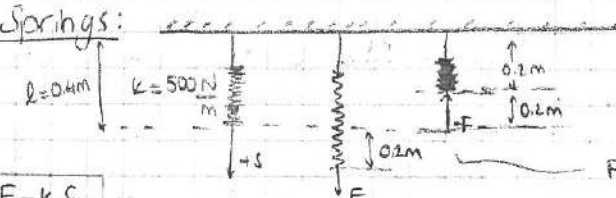
To apply the equation of equilibrium, we must account for all the known and unknown forces ($\sum \vec{F}$) which act on the particle. The best way to do this is to draw the particle's free-body diagram.



$$\sum \vec{F} = 0 \Rightarrow \begin{cases} \sum F_x = 0 \\ \sum F_y = 0 \end{cases}$$

$$\Rightarrow \sum F_x \hat{i} + \sum F_y \hat{j} = 0$$

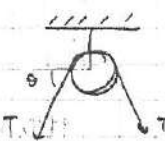
(a) Springs:



$$F = kS$$

$$F = kS = (500 \frac{\text{N}}{\text{m}})(0.2\text{m}) = -100\text{N}$$

Cables & Pulleys



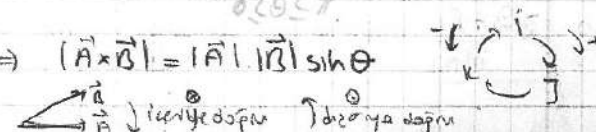
Cables (or cords) are assumed to have negligible weight and they can not stretch.

Cross Product

The cross product of two vectors $\vec{A}$ and $\vec{B}$ yields the vector $\vec{C}$ which is written

$$\vec{C} = \vec{A} \times \vec{B} \Rightarrow |\vec{A} \times \vec{B}| = |\vec{A}| |\vec{B}| \sin \theta$$

Area = $|\vec{A} \times \vec{B}|$



Method of Moments:

Moment of a force with respect to a point:

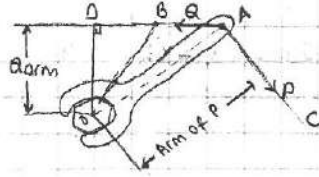


Fig.

$$M_O = -P \cdot (\text{Arm of } P)$$

$$M_O = Q \cdot (\text{Arm of } Q)$$

The wrench shown in Fig. to the handle of which two equal forces P and Q are applied as shown. The effectiveness or importance of a force, as regards its tendency to produce rotation of a body about a fixed point, is called the moment of the force with respect to that point.

✓ Ex:

Two smooth spheres, each of radius r and weight Q , rest in a horizontal channel having vertical walls, the distance between which is b (see Fig). Find the pressures exerted on the walls and floor at the points of contact A, B and D. The following data are given: $r = 10 \text{ cm}$, $b = 36 \text{ cm}$, $Q = 100 \text{ kg}$.

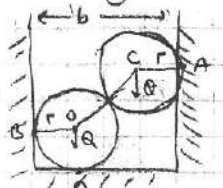
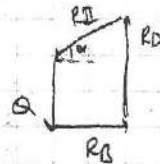
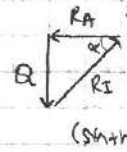


Fig.



$$\sum F_x = 0$$

$$\sum F_y = 0$$



$$\sum F_x = 0$$

$$\sum F_y = 0$$

(Synthesis)

action-reaction

$$R_I = R_{II}$$

II

$$R_I \cos \alpha - R_A = 0 \Rightarrow R_A = R_I \cos \alpha \Rightarrow R_A = \frac{Q}{\sin \alpha} \cos \alpha = Q \cot \alpha$$

$$-Q + R_I \sin \alpha = 0 \Rightarrow R_I = \frac{Q}{\sin \alpha}$$

11.10.16 (Sol.)

⑫

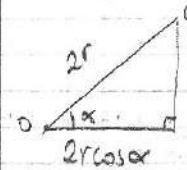
$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$-R_I \cos \alpha + R_B = 0 \Rightarrow R_B = R_I \cos \alpha \Rightarrow R_B = Q \cot \alpha$$

$$-R_I \sin \alpha - Q + R_D = 0 \Rightarrow R_D = Q + Q = 2Q$$

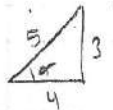
(uzunluğunu buluyoruz)



$$2r + 2r \cos \alpha = b$$

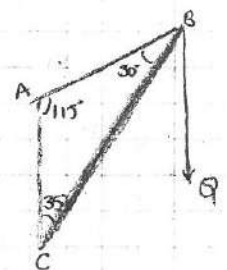
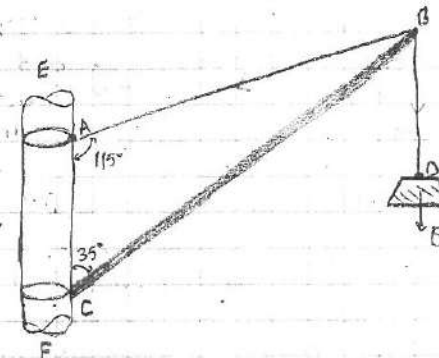
$$\cos \alpha = \frac{b - 2r}{2r} = \frac{b}{2r} - 1$$

$$\cos \alpha = \frac{36}{20} - 1 = \frac{4}{5} \Rightarrow$$



✓ Ex:

A weight $Q = 500 \text{ kg}$ hanging on a cable BD is supported at point B by a cable AB and a boom BC which is hinged at C (see Fig.1). Neglecting the weights of the cable and boom and assuming an ideal hinge at C, determine the forces transmitted to the mast at points A and C. The angles of $\triangle ABC$ are indicated in the figure.



by using sine theorem;

$$\frac{R_{CB}}{\sin 115^\circ} = \frac{Q}{\sin 30^\circ} = \frac{R_{AB}}{\sin 35^\circ}$$

$$R_{CB} = \frac{\sin 115^\circ}{\sin 30^\circ} \cdot Q$$

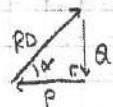
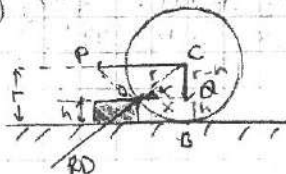
$$R_{AB} = \frac{\sin 35^\circ}{\sin 30^\circ} \cdot Q$$

Ex:

(13)

Determine the magnitude of a horizontal force P applied at the center C of a roller of weight Q and radius r which will be necessary to pull it over a curb.

(indimensional)



$$\sum F_x = 0; \sum F_y = 0$$

$$R \cos \alpha - P = 0 \Rightarrow P = R \cos \alpha \Rightarrow P = \frac{Q}{\sin \alpha} \cos \alpha = Q \cot \alpha$$

$$R \sin \alpha - Q = 0 \Rightarrow R = \frac{Q}{\sin \alpha}$$

$$\sin \alpha = \frac{r-h}{r}$$

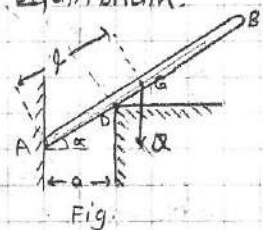
$$\cos \alpha = \frac{x}{r} \Rightarrow \cos \alpha = \frac{\sqrt{r^2 - (r-h)^2}}{r}$$

$$r^2 = x^2 + (r-h)^2 \Rightarrow x = \sqrt{r^2 - (r-h)^2}$$

$$Q \cdot \frac{\sqrt{r^2 - (r-h)^2}}{r} \cdot \frac{r}{r-h} \Rightarrow P = Q \cdot \frac{\sqrt{r^2 - (r-h)^2}}{r-h}$$

HW:

A slender prismatic bar AB of weight Q and length $2l$ rests on a very small frictionless roller at D, and against a smooth vertical wall at A, as shown in Fig. Find the angle α that the bar must make with the horizontal in the condition of equilibrium.



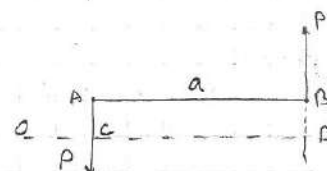
Homework deadline: 18.10.16

$$(Ans: \cos \alpha = \frac{3}{\sqrt{a}} \frac{l}{l})$$

Theorem of Varignon: The moment of the resultant of two concurred forces with respect to a center in their plane is equal to the algebraic sum of the moments of the components with respect to the same center.

(14)

Two Equal Parallel Forces Acting on Opposite Directions:



A system of two equal parallel forces acting in opposite directions can not be reduced to one resultant force.

$$M_o = P \cdot |OD| - P \cdot |OC| = P(|OD| - |OC|) = P \cdot a$$

Equilibrium:

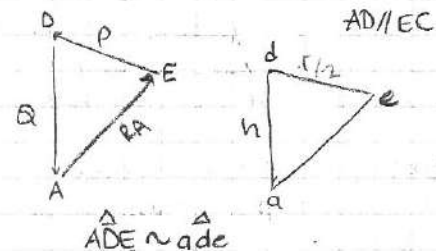
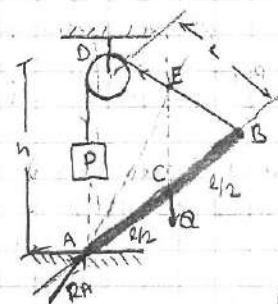
$\vec{R} = 0$

$$\begin{aligned} \textcircled{1} \sum F_x &= 0 \\ \sum F_y &= 0 \end{aligned} \quad \textcircled{2} \sum M_o = 0$$

13.10.16. Persimbe

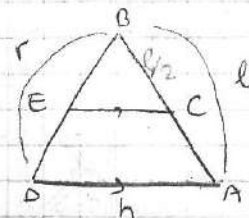
au d'magnd hantimili.

A prismatic bar AB of weight Q and length l is hinged at A and supported at B by a string that passes over a pulley D and carries a load P at its free end (see Fig.) Assuming that the distance h between the hinge A and the pulley D is larger than the length l of the bar, find the configuration of equilibrium of the system as defined by the ratio of lengths r/h .



$$\frac{|AD|}{|ad|} = \frac{|AE|}{|ae|} = \frac{|DE|}{|de|} \Rightarrow \frac{Q}{h} = \frac{RA}{|ae|} = \frac{P}{r/2}$$

$$\Rightarrow \frac{P}{Q} = \frac{r}{2h}$$



$$h - l < r < h + l$$

$$\frac{|EB|}{r} = \frac{l/2}{l} \Rightarrow |EB| = \frac{r}{2} \Rightarrow |ED| = \frac{r}{2}$$

15

$$\frac{h-l}{2h} < \frac{r}{2h} < \frac{h+l}{2h} \Rightarrow \frac{h-l}{2h} < \frac{P}{A} < \frac{h+l}{2h}$$

Plane Trusses: Method of joints:

A plane truss is defined as a system of bars, all lying in one plane and joined together at their ends in such way as to form a rigid framework.

It is usual practice in the analysis of trusses to make the following simplifying assumptions:

güçsüz
uzaylılar
sırtları
yapıtlar
kuvvetler
aynı düzlem
değirmi

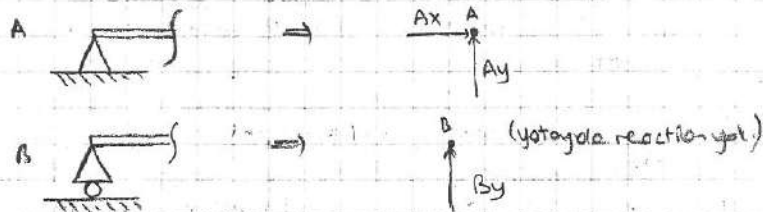
① that the bars are connected at their ends by frictionless hinges;

② that the axes of all bars lie in one plane; called the middle plane of the truss; and

③ that all forces acting on the truss are applied at the hinges only and also lie in the plane of truss.

It follows that the reactions which each bar of the truss exerts on the hinges at its two ends can be represented by two equal, opposite, and collinear forces which have for their common line of action the axis of the bar.

roller
or other
equivalent
reaction
support
(yörüngeli)
düzlemde
kuvvetler
aynı düzlem
değirmi



✓EX:

Determine the analytically the axial forces in the bars of the plane truss supported and loaded as shown in Fig ①

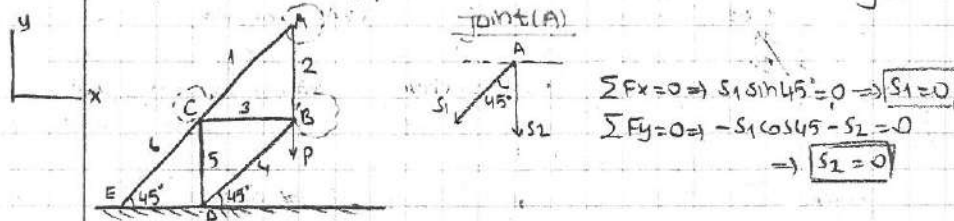


Fig ①

Joint (A): $S_1 \sin 45^\circ = 0 \Rightarrow S_1 = 0$
 $-S_1 \cos 45^\circ - S_2 = 0 \Rightarrow S_2 = 0$

Joint (B): $S_3 = 0$
 $-S_4 \cos 45^\circ - P + S_2 = 0 \Rightarrow S_4 = -\sqrt{2}P$ (compression)

compression
tension

16

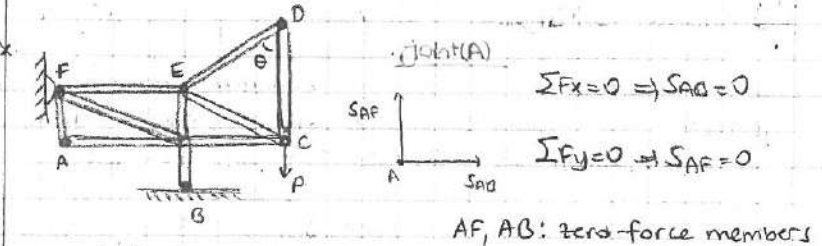
$$S_4 = \sqrt{2}P \text{ (compression) da yataklık.}$$

(1): $S_3 = -S_4 \cos 45^\circ = P$ (tension)

Joint (C): $\sum F_x = 0 \Rightarrow S_3 - S_6 \sin 45^\circ = 0 \Rightarrow S_6 = \frac{S_3}{\sin 45^\circ} = \sqrt{2}P$ (tension)
 $\sum F_y = 0 \Rightarrow -S_6 \cos 45^\circ - S_5 = 0 \Rightarrow S_5 = -S_6 \cos 45^\circ = -P$ (compression)

Zero Force Members:

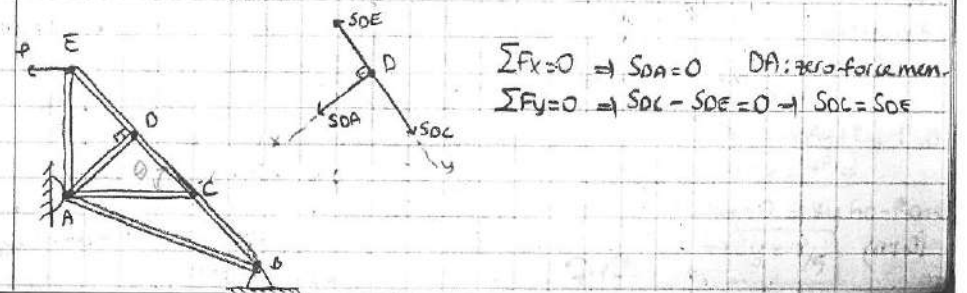
If only two members form a truss joint and no external load or support reaction is applied to the joint, the two members must be zero-force members.



Figure

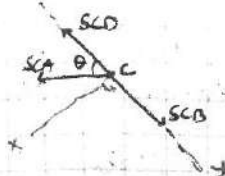
Joint (D): $\sum F_x = 0 \Rightarrow -S_{DE} \sin \theta = 0 \Rightarrow S_{DE} = 0$
 $\sum F_y = 0 \Rightarrow -S_{DE} \cos \theta - S_{DC} = 0 \Rightarrow S_{DC} = 0$
 (cos $\theta \neq 0$)
 DC, DE: zero-force members

If three members form a truss joint for which two of the members are collinear, the third member is a zero-force member provided no external force or support reaction is applied to the joint.



$\sum F_x = 0 \Rightarrow S_{DA} = 0$ DA: zero-force member
 $\sum F_y = 0 \Rightarrow S_{DC} - S_{DE} = 0 \Rightarrow S_{DC} = S_{DE}$

(17)



$$\sum F_y = 0 \Rightarrow S_{CB} - S_{CD} - S_{CA} \cos \theta = 0$$

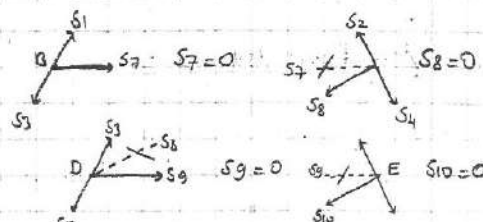
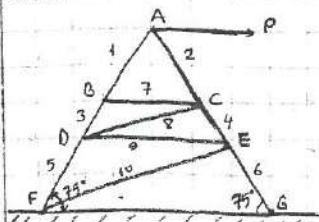
$$\Rightarrow S_{CB} = S_{CD}$$

$$\sum F_x = 0 \Rightarrow S_{CA} \sin \theta = 0 \Rightarrow (\sin \theta \neq 0) S_{CA} = 0$$

CA: zero-force member

EX:

Determine the force produced in each bar of the tower shown in Fig. due to a horizontal force P applied at the top as shown:



zero-force members

$$S_7 = S_8 = S_9 = S_{10} = 0$$

Joint (A):



$$\sum F_x = 0 \Rightarrow -S_1 \cos 75^\circ + S_2 \cos 75^\circ + P = 0$$

$$\Rightarrow S_1 = \frac{S_2 \cos 75^\circ + P}{\cos 75^\circ} = \frac{S_2 + P}{\cos 75^\circ}$$

$$\sum F_y = 0 \Rightarrow -S_1 \sin 75^\circ - S_2 \sin 75^\circ = 0$$

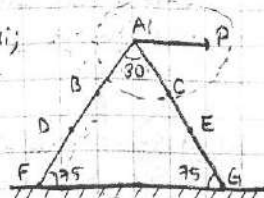
$$\Rightarrow S_1 = -S_2$$

$$S_1 = \frac{P}{2 \cos 75^\circ} \text{ (tension)}$$

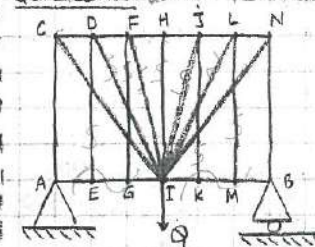
$$-2S_2 = \frac{P}{\cos 75^\circ} \Rightarrow S_2 = -\frac{P}{2 \cos 75^\circ} \text{ (compression)}$$

$$S_1 = S_3 = S_5 = \frac{P}{2 \cos 75^\circ} \text{ (tension)} \quad S_2 = S_4 = S_6 = -\frac{P}{2 \cos 75^\circ} \text{ (compression)}$$

Sevkin son hatı;
bayle duşunerek
iyi qatıyordun



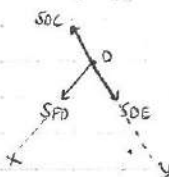
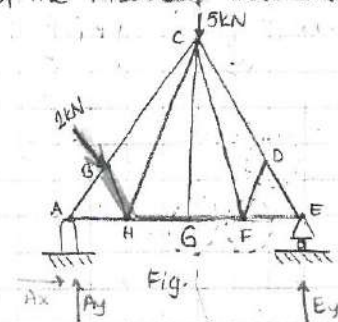
Quiz sorusu (Quiz-1) 22.10.2018



Find the zero-force members for the given plane truss.

EX:

(18) Using the method of joints, determine all the zero-force members of the truss shown in Fig. Assume all joints are pin connected



$$\sum F_x = 0 \quad S_{DF} = 0$$

$$\sum F_y = 0 \Rightarrow S_{DE} - S_{DC} = 0$$

$$S_{DE} = S_{DC}$$

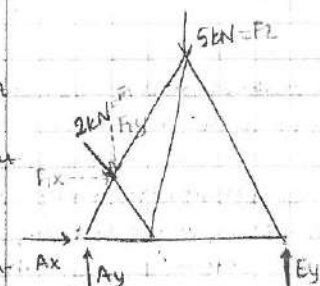
$$\sum F_x = 0 \Rightarrow -S_{FC} \sin \theta - S_{FG} + S_{FE} = 0$$

$$\sum F_y = 0 \Rightarrow S_{FC} \cos \theta = 0$$

$$\cos \theta \neq 0 \Rightarrow S_{FC} = 0$$

$$\theta \neq 90^\circ$$

Fixed support: A_x, A_y
Roller support: E_y



$$\sum F_x = 0 \Rightarrow A_x + F_{ix} = 0 \Rightarrow A_x = -F_{ix}$$

$$\sum F_y = 0 \Rightarrow A_y + E_y - F_{iy} - 5kN = 0 \Rightarrow F_{iy} = A_y + E_y - 5kN$$

$$\sum M_o = 0 \quad \sum M_A = 0 \rightarrow E_y \quad \sum M_E = 0 \rightarrow A_y$$

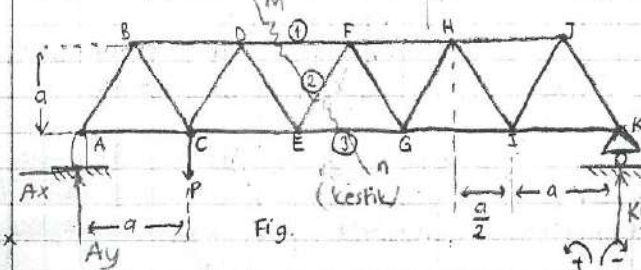
$$F_i = \sqrt{F_{ix}^2 + F_{iy}^2}$$

Plane trusses: Method of sections (Ritter)

We shall consider now another method of calculating the axial forces in the bars of a truss, the use of which makes it possible to determine the force in some chosen bar without going through successive considerations of the conditions of equilibrium of all hinges of the truss.

EX:

Referring to the truss shown in Fig., let us assume that it is required to determine the forces in the bars 1, 2, 3, due to a vertical load P applied as shown. Also, find the support reactions.

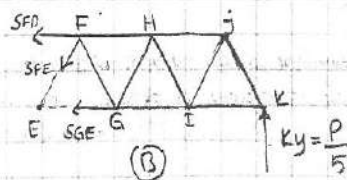
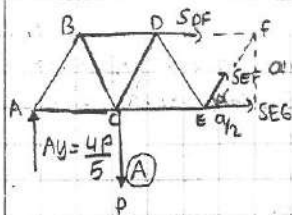


$$\sum F_x = 0 \Rightarrow A_x = 0$$

$$\sum F_y = 0 \Rightarrow A_y + K_y - P = 0 \Rightarrow A_y = \frac{4P}{5}$$

$$\sum M_A = 0 \Rightarrow K_y \cdot 5a - P \cdot a = 0 \Rightarrow K_y = \frac{P}{5} \quad (a \neq 0)$$

19



OR

2nd solution: $\sum M_E = 0 \Rightarrow$

$$-S_{DF}(a) - (4P/5)(2a) + P(a) = 0$$

$$S_{DF} = -3P/5 \text{ (compression)}$$

from (A)

$$\sum F_x = 0 \Rightarrow S_{DF} + S_{EG} + S_{EF} \cos \alpha = 0$$

$$\sum F_y = 0 \Rightarrow \frac{4P}{5} - P + S_{EF} \sin \alpha = 0 \Rightarrow S_{EF} = \frac{P}{5 \sin \alpha}$$

$$\sum M_F = 0 \Rightarrow -\frac{4P}{5}(2a + \frac{a}{2}) + P(a + \frac{a}{2}) + S_{EG}a = 0$$

$$\sum F_y = 0 \Rightarrow S_{EF} \sin \alpha + \frac{4P}{5} - P = 0 \Rightarrow S_{EF} = \frac{P}{5 \sin \alpha} = \frac{P}{5} \cdot \frac{2}{\sqrt{5}} = \frac{2P}{5\sqrt{5}} \text{ (tension)}$$

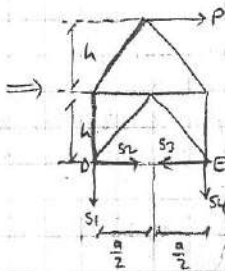
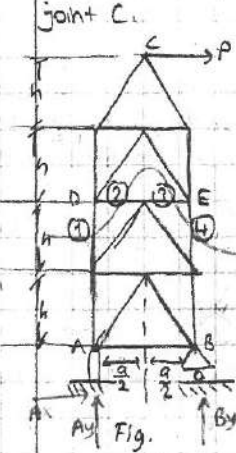
Remark: It consists essentially in the isolation of a portion of the truss by a section in such a way as to cause those internal forces that we wish to evaluate to become external forces on the isolated free body. The success or failure of the method rests entirely upon the choice of section. In general, a section should cut only three bars, since only three unknowns can be determined from three equations of equilibrium ($\sum F_x = 0, \sum F_y = 0, \sum M = 0$). However, there are special cases where we may successfully cut more than three bars.

Helodia!

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vor. Bunt
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ex. kaly
sewida
ayimay
kaly inqim

✓ Ex:

Determine the axial forces produced in the bars 1 and 4 of the tower shown in Fig. due to a horizontal force P applied at the joint C.



$$\sum F_x = 0 \Rightarrow S_2 + P - S_3 = 0$$

$$\Rightarrow S_2 - S_3 = -P$$

$$\sum F_y = 0 \Rightarrow -S_1 - S_4 = 0$$

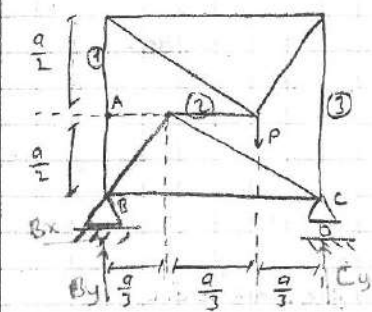
$$\Rightarrow S_1 = -S_4$$

$$\sum M_D = 0 \Rightarrow -S_4 a - P(2a) = 0 \Rightarrow S_4 = \left(\frac{-2h}{a} \right) P$$

(tension) $S_1 = \left(\frac{2h}{a} \right) P$ $\rightarrow$ compression

✓ Ex:

Determine the forces in bars 1, 2 and 3 of the plane truss supported and loaded as shown in Fig. (Ans: $S_1 = -P/3$; $S_2 = 0$; $S_3 = -2P/3$) (Also find support reactions)

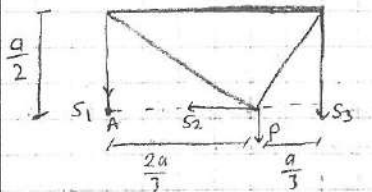


$$\sum F_x = 0 \Rightarrow B_x = 0$$

$$\sum F_y = 0 \Rightarrow B_y + C_y - P = 0 \Rightarrow B_y = \frac{P}{3}$$

$$\sum M_B = 0 \Rightarrow -2aP + aC_y = 0$$

$$\Rightarrow C_y = \frac{2P}{3}$$



$$\sum F_x = 0 \Rightarrow -S_2 = 0 \Rightarrow S_2 = 0$$

$$\sum F_y = 0 \Rightarrow -S_1 - P - S_3 = 0 \Rightarrow S_1 = -\frac{P}{3}$$

$$\sum M_A = 0 \Rightarrow -\frac{2a}{3}P - aS_3 = 0 \Rightarrow S_3 = -\frac{2P}{3}$$

Friction



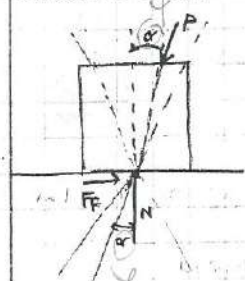
Whenever the surfaces at two bodies are in contact there will be limited amount of resistance to sliding between them, which is called friction. For given dry surfaces in contact, the result may be summarized briefly by the following laws of friction.

- 1) The total friction that can be developed is independent of the magnitude of the area of contact.
- 2) The total friction that can be developed is proportional to the normal force.
- 3) For low velocities of sliding the total friction that can be developed is practically indep. of the velocity, although the experiments show that the force F necessary to start sliding is greater than that necessary to maintain sliding. Friction may be expressed by $F = \mu N$, μ is coefficient of friction.

20.10.2016

(21)

If F is taken as the force necessary to start sliding, μ is called coefficient of static friction. If F is taken as the some what smaller force necessary to maintain sliding once it has been started, μ is called the coefficient of kinetic friction.



To see how to effect the reactions exerted by supporting surfaces, let's consider the simple case of a small block resting upon a horizontal plane.

$$\tan \alpha = \frac{F_f}{N}$$

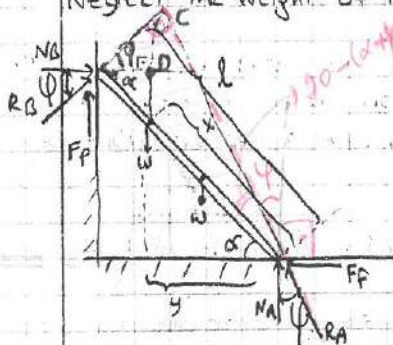
When we applied force P , makes a certain limiting angle α with the vertical such that sliding of the block impends, we have as before, as a condition of equilibrium that

$$\frac{F_f}{N} = \tan \alpha$$

Also, as a condition of impending sliding $\frac{F}{N} = \mu$. So, $\tan \alpha = \mu$, where " α " is angle of friction."

EX!

A ladder AB of length l is supported by a horizontal floor at A and by vertical wall at B and makes an angle α with the horizontal. Find the max. distance x up the ladder at which a man of weight w can stand without causing slipping to occur if the angle of friction between floor and ladder and between wall and the ladder is ϕ . Neglect the weight of the ladder itself.



$$\cos \alpha = \frac{y}{x} \quad y = x \cos \alpha$$

$$|BD| = l \cos \alpha - x \cos \alpha$$

$$x = l - |BD| \sec \alpha$$

$$|BD| = |BC| \cos \phi$$

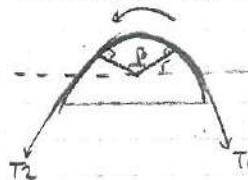
$$|BD| = \frac{|BC|}{\cos \phi} \cdot \cos \phi$$

$$x = l (1 - \cos(\alpha + \phi) \cdot \cos \phi \sec \alpha)$$

(22)

Frictional Forces on Flat Belts

$$T_2 = T_1 e^{\mu \phi} \quad \phi: \text{radian}$$

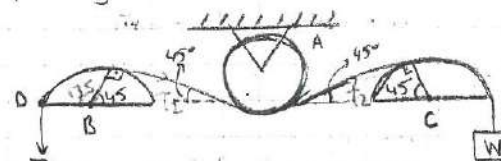


Where T_1, T_2 are belt tensions, T_1 opposes to the direction of motion of belt measured relative to the surface, while T_2 acts in the direction of the relative belt motion because $T_2 > T_1$. μ : coefficient of static and kinetic friction between belt and the surface of contact.

ϕ : angle of belt to surface contact, measured in radians.

EX!

The max. tension that can be developed in the cord shown in fig. is 500 N if the pulley at A is free to rotate and the coefficient of the static friction at the fixed drums B and C is $\mu = 0.25$. Determine the largest mass of cylinder that can be lifted by the cord.



$$W = mg \quad g = 9.81 \text{ m/s}^2$$

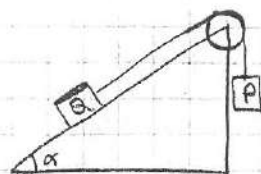
$$T = 500 \text{ N} = T_2 \cdot e^{(0.25) \cdot \frac{3\pi}{4}} \Rightarrow T_2 = 277.4 \text{ N}$$

$$277.4 = \frac{m \times 9.81}{W} e^{3\pi/16} \Rightarrow m = 15.7 \text{ kg}$$

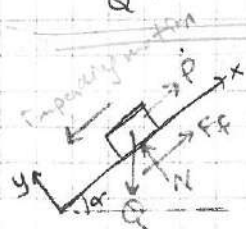
EX!

A block of weight Q on an incline plane and has attached to it a string that overruns the pulley and carries a weight P and its other end. If the coefficient of friction between the block Q and the inclined plane is μ , find the limiting values of the ratio P/Q consistent with equilibrium. Neglect friction in the pulley and assume that the angle of inclination α of the plane is greater than the angle of friction $\phi = \arctan \mu$. (ki hanna olun kayda baglayay)

(23)



case 1: $\frac{P}{Q} \ll 1 \Rightarrow P \ll Q$

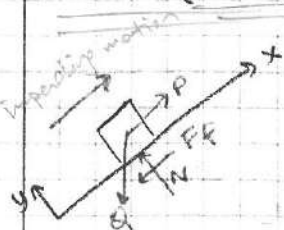


$$\sum F_x = 0 \Rightarrow F_f - Q \sin \alpha + P = 0$$

$$Q \sin \alpha = \mu N + P \Rightarrow Q \sin \alpha - \mu N = P$$

$$\sum F_y = 0 \Rightarrow N - Q \cos \alpha = 0 \Rightarrow N = Q \cos \alpha$$

case 2: $\frac{P}{Q} \gg 1 \Rightarrow P \gg Q$



$$\sum F_x = 0 \Rightarrow P - F_f - Q \sin \alpha$$

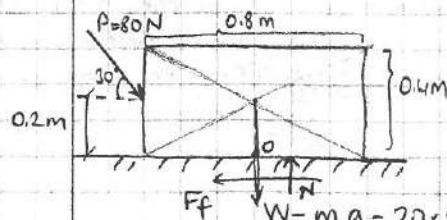
$$Q \sin \alpha + \mu N = P$$

$$\sum F_y = 0 \Rightarrow N - Q \cos \alpha \Rightarrow N = Q \cos \alpha$$

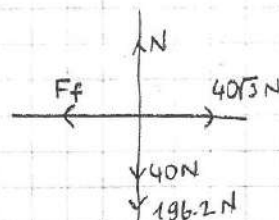
From case 1: $\frac{P}{Q} = \sin \alpha - \mu \cos \alpha \quad (1)$

From case 2: $\frac{P}{Q} = \sin \alpha + \mu \cos \alpha \quad (2)$

EX: The uniform crate has a mass of 20 kg. If a force $P=80\text{ N}$ is applied to the crate determine if it remains in equilibrium. The coefficient of static friction is $\mu=0.3$.



$$W = m \cdot g = 20 \times 9.81 = 196.2\text{ N}$$



(24)

$$\sum F_x = 0 \Rightarrow P \cos 30^\circ - F_f = 0 \Rightarrow F_f = 40\sqrt{3} = 69.3\text{ N} \quad (1)$$

$$\sum F_y = 0 \Rightarrow N - W - P \sin 30^\circ = 0 \Rightarrow N = 236.2\text{ N}$$

$$\sum M_o = 0 \Rightarrow N \cdot x - P \cos 30^\circ (0.2) + P \sin 30^\circ (0.4)$$

$$x = -0.00908\text{ m} \quad |x| < 0.4$$

$$F_f = \mu N = (0.3)(236.2) = 70.8\text{ N} \quad (2)$$

(1 < 2 so it is in equilibrium.)

Dahis system olur. (Dahis paranti)

$$(M_s)_{\min} = 0.2679491924$$

$$(M_s')_{\min} = 0.0893163975$$



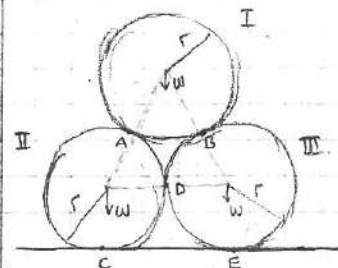
$$\mu_c = \mu_s'$$

$$F_c = F = (0.1339745962)W \quad F_c = \mu_c \cdot N_c \quad (11)$$

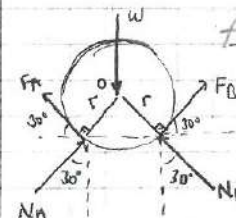
$$\text{from Eq. (8)}: N_c = N \cos 30^\circ + W + F \sin 30^\circ$$

$$N_c = \left(\frac{W}{2}\right) \frac{\sqrt{3}}{2} + W + (0.1339745962)W \left(\frac{1}{2}\right) = \frac{6W}{4} = 1.5W \quad (12)$$

$$\text{from Eq. (11)}: \mu_c = (\mu_s') = \frac{(0.1339745962)W}{(1.5)W} = 0.0893163975$$



The concrete pipes are stacked in the yard as shown in the left figure, determine the minimum coefficient of friction at each point of contact so that the pipe doesn't collapse.



from Eqs. (1) and (3): $N_A = N_B = N \quad (4)$

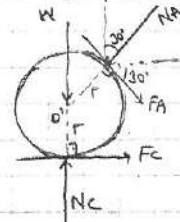
$$\rightarrow \sum F_x = 0 \Rightarrow N \sin 30^\circ - F_A \cos 30^\circ - N_B \sin 30^\circ + F_B \cos 30^\circ = 0 \quad (1)$$

$$\uparrow \sum F_y = 0 \Rightarrow F_A \sin 30^\circ + F_B \sin 30^\circ + N_A \cos 30^\circ + N_B \cos 30^\circ - W = 0 \quad (2)$$

$$\circlearrowleft \sum M_O = 0 \Rightarrow -F_A(r) + F_B(r) = 0 \Rightarrow F_B = F_A = F \quad (3)$$

$$2F \sin 30^\circ + N \cos 30^\circ + N \cos 30^\circ - W = 0$$

from Eqs. (2) and (3): $2F \sin 30^\circ + 2N \cos 30^\circ - W = 0 \quad (5)$



$$\sum M_O = 0 \Rightarrow -F_A(r) + F_C(r) = 0 \Rightarrow F_A = F_C = F \quad (6)$$

$$\sum F_x = 0 \Rightarrow F_C + F_A \cos 30^\circ - N \sin 30^\circ = 0 \quad (7)$$

$$\sum F_y = 0 \Rightarrow N \cos 30^\circ - W - F_A \sin 30^\circ = 0 \quad (8)$$

from Eqs. (6) and (7):

$$\mu_s = \frac{F}{N} = \frac{1}{2\sqrt{3}} = 0.2886751345$$

$$F + F \cos 30^\circ - N \sin 30^\circ = 0 \Rightarrow F = \frac{\sin 30^\circ}{1 + \cos 30^\circ} N \quad \Rightarrow \frac{F}{N} = \frac{1/2}{1 + \sqrt{3}/2}$$

$$N_c = N \cos 30^\circ + W + F \sin 30^\circ = N \cos 30^\circ + W + \frac{\sin^2 30^\circ}{1 + \cos 30^\circ} N$$

from Eq. (5):

$$(1): 2F \sin 30^\circ + 2N \cos 30^\circ = W \Rightarrow 2 \left(\frac{F}{N}\right) \sin 30^\circ = \frac{W}{N}$$

$$\Rightarrow (2 \cos 30^\circ) (0.2679) + 2 \cos 30^\circ = \frac{W}{N}$$

$$N = \frac{W}{(2 \cos 30^\circ) (0.2679) + 2 \cos 30^\circ} \approx 0.5W$$

Hotall?

from Eq. (5):

$$2F \sin 30^\circ + 2 \left(\frac{W}{2}\right) \cos 30^\circ - W = 0 \Rightarrow F = W (1 - \cos 30^\circ) \Rightarrow F = 0.1339745962W$$

(27)

$$\frac{F}{N} = \frac{\sin 30^\circ}{1 + \cos 30^\circ} = (\mu_s)_{\min} = 0.2679 \quad (F = \mu N) \text{ gerige Koppel}$$

$$(2): N_c = \left(\cos 30^\circ + \frac{\sin^2 30^\circ}{1 + \cos 30^\circ} \right) (0.5W) + W = W \left(1 + \left(\cos 30^\circ + \frac{\sin^2 30^\circ}{1 + \cos 30^\circ} \right) \right)$$

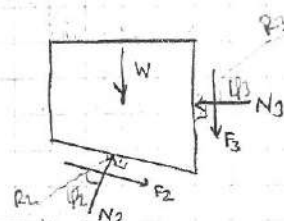
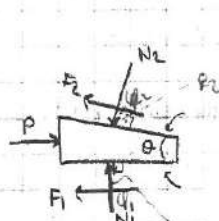
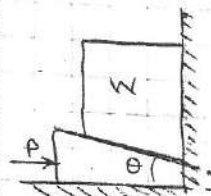
$$F = (\mu_s) N_c \Rightarrow (\mu_s) (1.5W) = \left(\frac{\sin 30^\circ}{1 + \cos 30^\circ} \right) \cdot 0.5W$$

$$\Rightarrow (\mu_s)_{\min} = \left(\frac{\sin 30^\circ}{1 + \cos 30^\circ} \right) \cdot \frac{1}{3} \approx 0.0893$$

$$(\mu_s)_{\min} = 0.2679; (\mu_s')_{\min} = 0.0893$$

Friction on Wedges:

A wedge is a simple machine that is often used to transform an applied force into much larger forces, directed at approximately right angles to the applied force. Wedges also can be used to slightly move or adjust heavy loads.



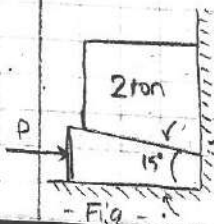
(a) impending motion

(b) FBD of wedge

(c) FBD of block

Ex!

To raise a heavy stone block weighting 2 ton, the arrangement shown in Fig. is used. What horizontal force P will it be necessary to apply to the wedge in order to raise the block if the coefficient of friction for all contiguous surfaces is $\mu = \frac{1}{4}$? Neglect the weight of the wedge.



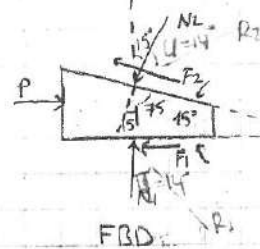
$$\mu = \frac{1}{4} \quad \tan \phi = \mu = \frac{F}{N}$$

$$\phi = \arctan \mu = \arctan(0.25) = 14^\circ$$

$$\phi = 14.03624347^\circ$$

(28)

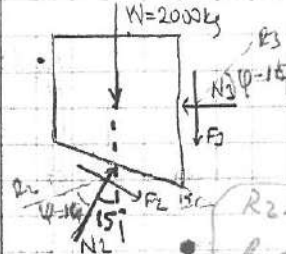
Equilibrium of wedge:



$$\sum F_x = 0 \Rightarrow -R_1 \sin 14^\circ + P - R_2 \sin(14^\circ + 15^\circ) = 0$$

$$\sum F_y = 0 \Rightarrow R_1 \cos 14^\circ - R_2 \cos(14^\circ + 15^\circ) = 0$$

Equilibrium of the block:



$$\sum F_x = 0 \Rightarrow R_2 \sin(14^\circ + 15^\circ) - R_3 \cos(14^\circ) = 0$$

$$\sum F_y = 0 \Rightarrow R_2 \cos(14^\circ + 15^\circ) - 2000 \text{ kg} - R_3 \sin(14^\circ) = 0$$

$$R_2 = \frac{2000 \text{ kg} + R_3 \sin(14^\circ)}{\cos(14^\circ + 15^\circ)}$$

$$(1): P = R_1 \sin 14^\circ + R_2 \sin 29^\circ = R_2 \left(\sin 14^\circ \cdot \frac{\cos 29^\circ}{\cos 14^\circ} + \sin 29^\circ \right) = R_2 \cdot \frac{\sin 43^\circ}{\cos 14^\circ}$$

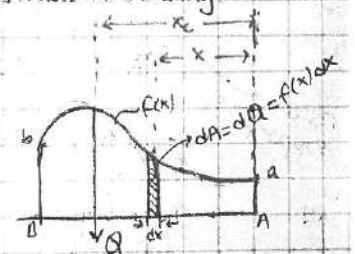
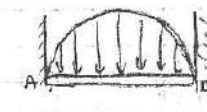
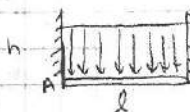
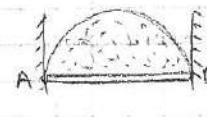
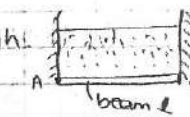
$$(2): R_1 = R_2 \cdot \frac{\cos 29^\circ}{\cos 14^\circ}$$

$$(3): R_3 = R_2 \cdot \frac{\sin 29^\circ}{\cos 14^\circ}$$

$$(4): R_3 = \frac{2000 \text{ kg} - R_2 \cos 29^\circ}{-\sin 14^\circ}$$

Reduction of Simple Distributed Loading:

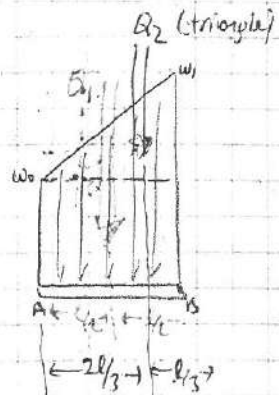
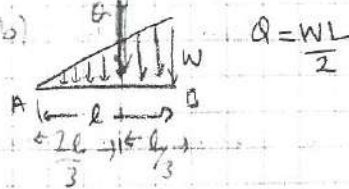
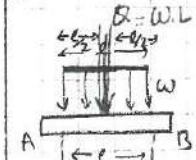
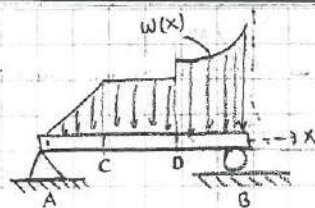
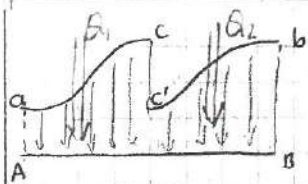
We have deal with distributed force such as water pressure against the face of dam, or ordinary gravity force, the weight of a sand on the floor of a storage container, are all distributed loading



$$Q = \int_A^B f(x) dx$$

$$x_c = \frac{\int_A^B x f(x) dx}{Q}$$

(29)

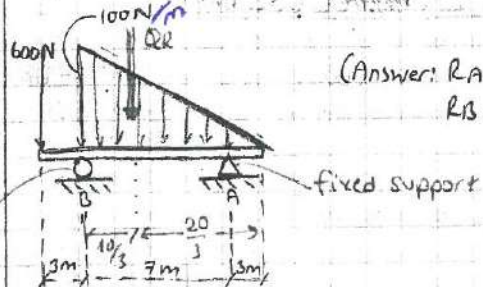


$$Q_1 = w_0 L$$

$$Q_2 = \frac{(w_1 - w_0) L}{2}$$

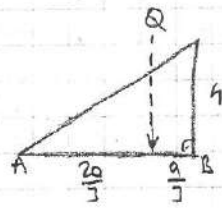
Ex!

Calculate the reactions at A and B for the beam loaded as shown in Fig. Neglect weight of the beam.



(Answer: $R_A = 19 \text{ N}$ (down)
 $R_B = 1119 \text{ N}$ (up))

01.11.2016 Sol.



$$Q_{\text{resultant}} = \frac{(100 \text{ N/m})(10 \text{ m})}{2} = 500 \text{ N}$$



$$\sum F_x = 0 \Rightarrow A_x = 0$$

$$\sum F_y = 0 \Rightarrow A_y + B_y - Q_2 - 600 \text{ N} = 0$$

$$A_y + B_y = 1100 \text{ N}$$

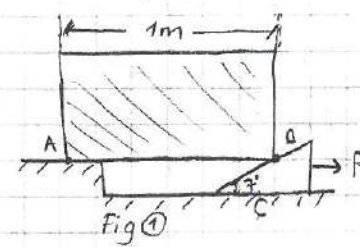
(30)

$$\sum M_A = 0 \Rightarrow -B_y(7 \text{ m}) + (600 \text{ N})(10 \text{ m}) + Q_2 \left(\frac{20}{3} - 3 \right) = 0$$

$$\Rightarrow B_y = \frac{6000 + (500)(11/3)}{7} \quad (B_y = 1119.05 \text{ N (up)})$$

$$A_y = 1100 \text{ N} - 1119.05 \text{ N} \quad (A_y = -19.05 \text{ N (down)})$$

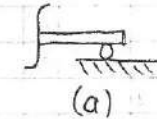
HW2 The uniform stone in Fig 1 has a mass of 500 kg and is held in the horizontal position using a wedge at B. If the coefficient of static friction is $\mu_s = 0.3$ at the surfaces of contact, determine the minimum force P needed to remove the wedge. Assume that the stone does not slip at A.



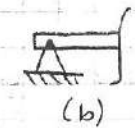
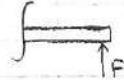
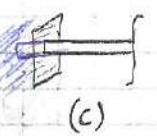
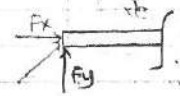
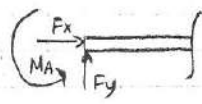
$$(P = 1154.9 \text{ N})$$

Deadline: 08.11.2016.

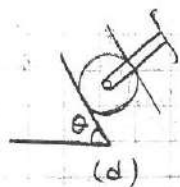
Support Reactions



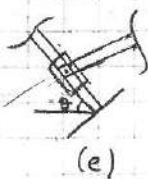
Roller

Pin (or hinge)
(fixed support)built in support
(or fixed support)

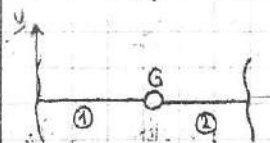
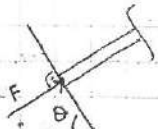
(31)



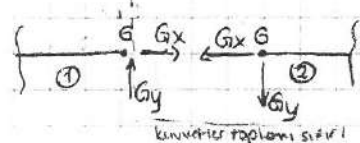
roller or pin confined
smooth slot



member pin connected
to collar on smooth rod.



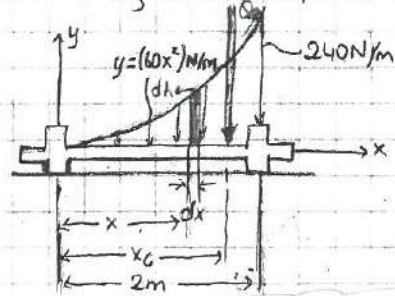
Gerber support



(f)

Ex:

Determine the magnitude and location of the equivalent force acting on the shaft shown in Fig. 1.



$$dA = y \cdot dx = (60x^2) N/m \cdot dx$$

$$A = Q_R = \int_0^{2m} (60x^2) dx = 60x^3 \Big|_0^{2m} = 60 \cdot \frac{8}{3} = 160 N$$

$$\sum M_O = -x_1 dA_1 + x_2 dA_2 + \dots - x_n dA_n$$

$$\sum M_O = -Q_R \cdot x_c$$

$$\sum M_O = Q_R \cdot x_c$$

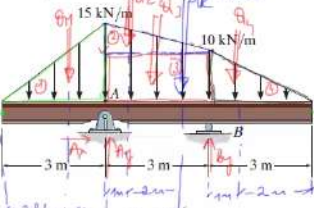
$$dA = dA_1 + dA_2 + dA_3 + \dots + dA_n$$

$$Q_R \cdot x_c = \int_0^{2m} x \cdot dA \Rightarrow x_c = \frac{\int_0^{2m} x y dx}{Q_R} = \frac{\int_0^{2m} x (60x^2) N/m dx}{160 N}$$



$$x_c = \frac{\int_0^{2m} 60x^3 dx}{160 N} = \frac{60 \frac{x^4}{4} \Big|_0^{2m}}{160 N} = \frac{60 \cdot \frac{16}{4}}{160 N} = 1.5 m$$

Ex. Replace the distributed loading with an equivalent resultant force, and specify its location on the beam measured from point A. And also find the support reactions at A and B.



$$Q_1 = \frac{(3m)(75 kN/m)}{2} = \frac{45}{2} kN$$

$$Q_2 = \frac{(15 kN - 75 kN) \cdot (3m)}{2} = \frac{15}{2} kN$$

$$Q_3 = (10 kN/m) \cdot (3m) = 30 kN$$

$$Q_4 = (60 kN/m) \cdot (3m) = 180 kN$$

$$Q_R = Q_1 + Q_2 + Q_3 + Q_4$$

$$= \frac{45}{2} + \frac{15}{2} + 30 + 180 = 240 kN$$

$$Q_R = 240 kN$$

by using Varignon's theorem:

$$-Q_R \cdot x_c = Q_1 \cdot (1m) - Q_2 \cdot (1m) - Q_3 \cdot (1.5m) - Q_4 \cdot (6m)$$

$$(-240 kN) \cdot (x_c) = \left(\frac{45}{2} - \frac{15}{2} - 45 - 180 \right) kN \cdot m$$

$$-240 \cdot x_c = -240 m \Rightarrow x_c = \frac{240}{240} m = 1 m$$

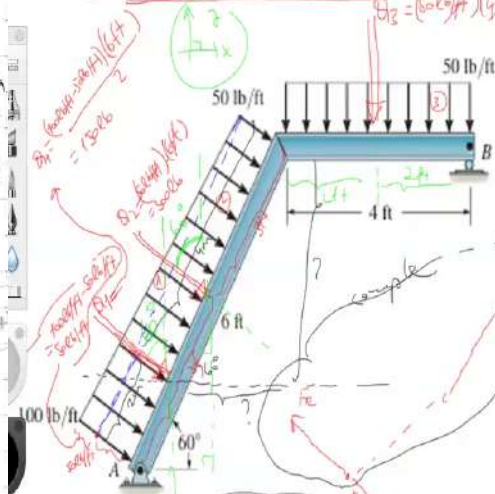
$$\sum F_x = 0 \Rightarrow A_x = 0$$

$$\sum F_y = 0 \Rightarrow A_y + B_y - Q_1 - Q_2 - Q_3 - Q_4 = 0 \Rightarrow A_y + B_y = 240 kN = Q_R$$

$$\sum M_A = 0 \Rightarrow Q_1 \cdot (1m) - Q_2 \cdot (1m) - Q_3 \cdot (1.5m) - Q_4 \cdot (6m) + B_y \cdot (3m) = 0 \Rightarrow B_y = \frac{90 kN \cdot m}{3m} = 30 kN$$

$$A_y = B_y = 30 kN$$

Ex. Replace the loading by an equivalent resultant force and couple moment acting at point B.



$$\sum F_x = Q_1 \cdot \sin 60^\circ + Q_2 \cdot \sin 60^\circ = (50 kN) \left(\frac{\sqrt{3}}{2} \right) + (100 kN) \left(\frac{\sqrt{3}}{2} \right) = 250 \sqrt{3} kN$$

$$F_x = 250 \sqrt{3} kN$$

$$\sum F_y = -Q_1 \cdot \cos 60^\circ - Q_2 \cdot \cos 60^\circ$$

$$= -(50 kN) \left(\frac{1}{2} \right) - (100 kN) \left(\frac{1}{2} \right) = -75 kN$$

$$F_y = -75 kN$$

$$F_R = \sqrt{F_x^2 + F_y^2} = \sqrt{(250 \sqrt{3})^2 + (-75)^2} = 576.6277 kN \approx 577 kN$$

$$\tan \theta = \left| \frac{F_y}{F_x} \right| = \frac{75}{250 \sqrt{3}} = 1.0906$$

$$\theta = \tan^{-1}(1.0906) = 47.980^\circ$$

$$\sum M_B = M_B + (Q_1 \cdot \sin 60^\circ) \cdot (4 \sin 60^\circ) + (Q_2 \cdot \cos 60^\circ) \cdot (4 \cos 60^\circ) + \dots$$

33

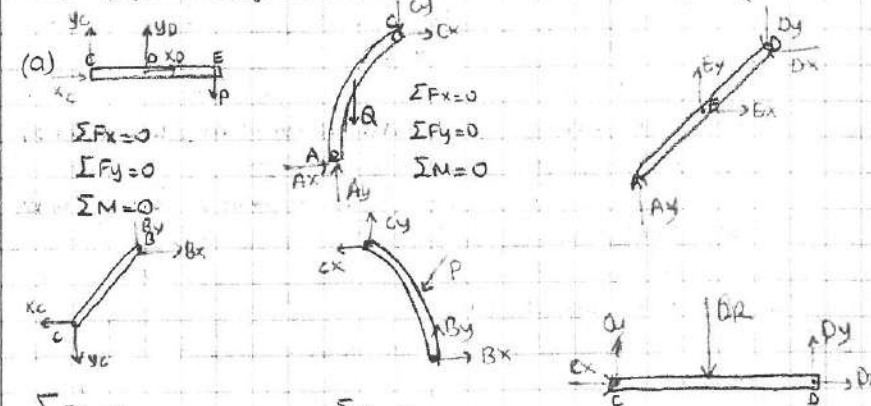
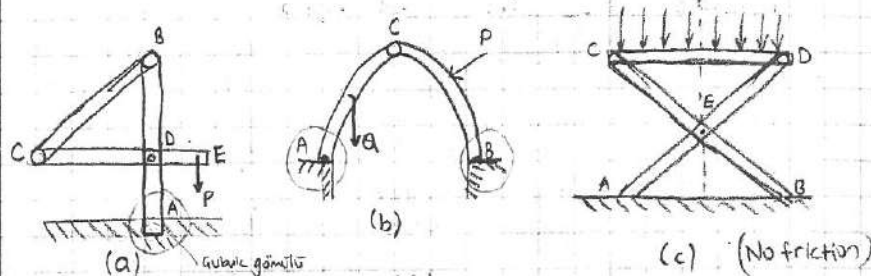
Frames and Machines

03.11.2016

Frames are generally stationary and are used to support loads, whereas machines contain moving parts and are designed to transmit and alter the effect of forces.

Free Body Diagram:

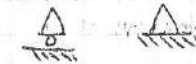
Isolate each part by drawing its outlined shape. Then show all the forces and/or couple moment that act on the part. Make sure to label or identify each known and unknown force and couple moment with reference to an established x-y coordinate system.



$$\begin{aligned}\sum F_x &= 0 \\ \sum F_y &= 0 \\ \sum M &= 0\end{aligned}$$

$$\begin{aligned}\sum F_x &= 0 \\ \sum F_y &= 0 \\ \sum M &= 0\end{aligned}$$

Frames

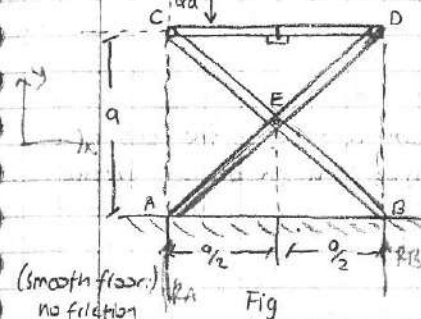


$$\begin{aligned}\sum F_x &= 0 \\ \sum F_y &= 0 \\ \sum M &= 0\end{aligned}$$

Ex:

34

A folding campstool rests upon a smooth horizontal floor and is loaded as shown in Fig. Determine the magnitude of the shear force on the pin E and the position of the load P on the bar CD (that is, the value of α) to make this force a maximum.



for the entire frame:

$$\sum F_x = 0$$

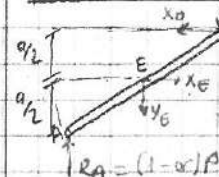
$$\sum F_y = 0 \Rightarrow R_A + R_B - P = 0$$

$$\sum M_A = 0 \text{ (or } \sum M_B = 0) \Rightarrow R_B \cdot a - P \cdot \alpha a = 0$$

$$\Rightarrow R_B = \alpha P$$

$$\alpha P + R_A - P = 0 \Rightarrow R_A = (1 - \alpha)P$$

for the AED member



$$\sum F_x = 0 \Rightarrow -x_D + x_E = 0$$

$$\Rightarrow x_D = x_E$$

$$\sum F_y = 0 \Rightarrow (1 - \alpha)P - y_E - y_D = 0$$

$$\sum M_E = 0 \Rightarrow -R_A \left(\frac{a}{2}\right) + x_D \left(\frac{a}{2}\right) - y_D \left(\frac{a}{2}\right) = 0$$

$$x_D - y_D = (1 - \alpha)P$$

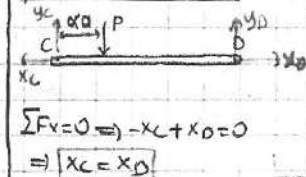
$$E: \begin{cases} x_E = P \\ y_E = (1 - 2\alpha)P \end{cases}$$

$$R_E = \sqrt{x_E^2 + y_E^2} = P \sqrt{1 + (1 - 2\alpha)^2}$$

$(R_E)_{\max} = ?$ $\alpha = ?$ differentiate R_E w.r.t α :

$$(R_E)' = \frac{2(1 - 2\alpha)(-2)}{2\sqrt{1 + (1 - 2\alpha)^2}} = 0 \Rightarrow -4 + 8\alpha = 0 \Rightarrow \alpha = 1/2$$

for the CD member



$$\sum F_x = 0 \Rightarrow -x_C + x_D = 0$$

$$\Rightarrow x_C = x_D$$

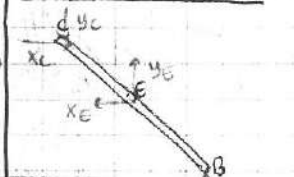
$$\sum F_y = 0 \Rightarrow y_C + y_D - P = 0$$

$$y_C = (1 - \alpha)P$$

$$\sum M_C = 0 \Rightarrow -P(\alpha a) + y_D(a) = 0$$

$$y_D = \alpha P$$

for the BEC member



$$\sum F_x = 0 \Rightarrow x_C - x_E = 0$$

$$\sum F_y = 0 \Rightarrow R_B + y_E - y_C = 0$$

$$\sum M_E = 0 \Rightarrow$$

$$0 \leq \alpha \leq 1$$

35

$$R_E = P \sqrt{1 + (1-2\alpha)^2}$$

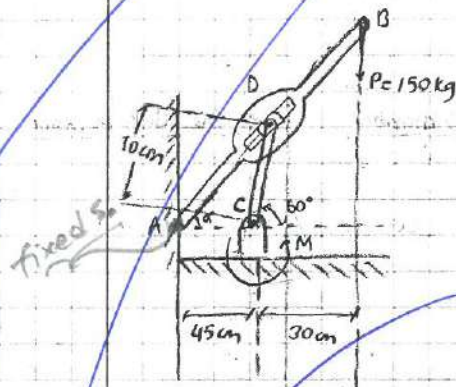
$$\alpha = 1/2 \Rightarrow R_E = P$$

$$\alpha = 0 \text{ (P on C)} \Rightarrow R_E = \sqrt{2} P$$

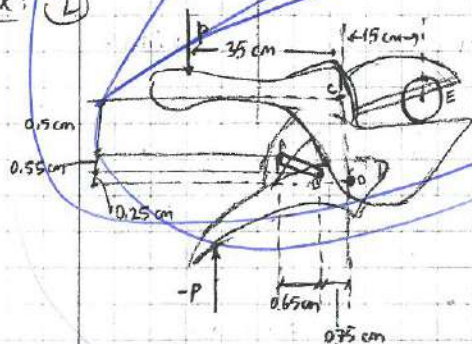
$$\alpha = 1 \text{ (P on D)} \Rightarrow R_E = \sqrt{2} P$$

$$(R_E)_{\max} = \sqrt{2} P$$

Ex: 1 Determine the couple M that must be applied to the crank CD to hold the mechanism in equilibrium. The block at D is pinned to the crank CD and is free to slide in a slot cut in member AB.



Ex: 2

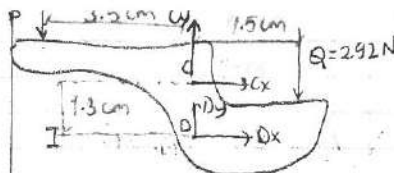


The compound-lever pruning shears shown can be adjusted by placing pin A at various ratchet positions on blade ACE. Knowing that 292 N vertical forces are required to complete the pruning of a twig, determine the magnitude P of the forces that must be applied to the handles when the shears are adjusted as shown.

Solution of example:

ortak noktalarda büyükm her parçacıkta ve zitiyönü Gitiyörüm gitiyönü.

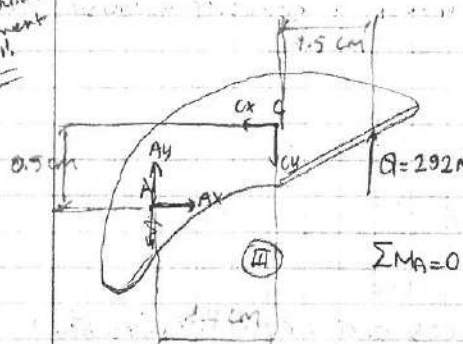
36



$$\sum F_x = 0 \Rightarrow C_x + D_x = 0 \quad (1)$$

$$\sum F_y = 0 \Rightarrow C_y + D_y - P - 292 \text{ N} = 0 \quad (2)$$

$$\sum M_O = 0 \Rightarrow P(3.5 \text{ cm}) - C_x(1.3 \text{ cm}) - (292 \text{ N})(1.5 \text{ cm}) = 0 \quad (3)$$

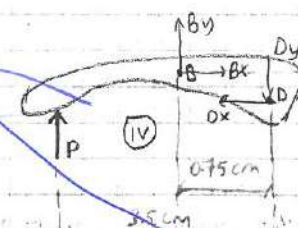


$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum M_{A/C} = 0$$

$$\sum M_A = 0 \Rightarrow (292 \text{ N})(2.9 \text{ cm}) + C_y(1.4 \text{ cm}) + C_x(0.5 \text{ cm}) = 0 \quad (4)$$

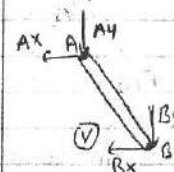


$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum M_{B/D} = 0$$

$$\sum M_B = 0 \Rightarrow -P(3.5 - 0.75 \text{ cm}) - D_x(0.9 \text{ cm}) - D_y(0.75 \text{ cm}) = 0 \quad (5)$$



$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum M_{A/B} = 0$$

Cx	Cy	Dx	Dy	P
1	0	1	0	0 = 0

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 \\ - & - & - & - & - \end{bmatrix} \begin{bmatrix} C_x \\ C_y \\ D_x \\ D_y \\ P \end{bmatrix} = \begin{bmatrix} 0 \\ 292 \\ 1 \end{bmatrix}$$

$$C_x = -260 \text{ N}$$

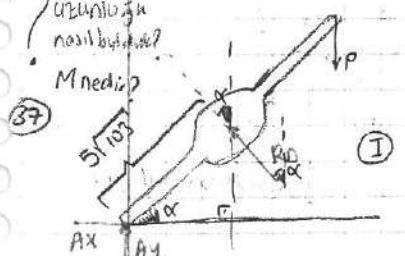
$$C_y = 512 \text{ N}$$

$$D_x = 260 \text{ N}$$

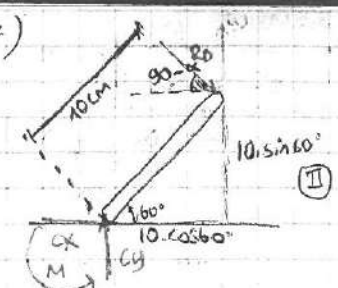
$$D_y = -191.428 \text{ lb}$$

$$P = 28.5714 \text{ lb}$$

37 solution of (1) (konsistentde Rodje birzy yak)



$$\begin{aligned} \sum F_x = 0 &\Rightarrow X_A - R \sin \alpha = 0 \quad (1) \\ \sum F_y = 0 &\Rightarrow Y_A + R \cos \alpha - P = 0 \quad (2) \\ \sum M_A = 0 &\Rightarrow R D (5\sqrt{103}) - P (75 \text{ cm}) = 0 \end{aligned}$$



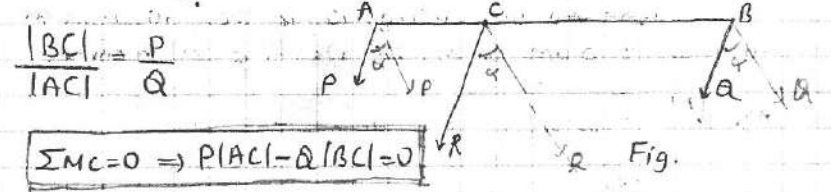
$$\begin{aligned} \sum F_x = 0 &\Rightarrow R D \cos(90 - \alpha) + C_x = 0 \\ \sum F_y = 0 &\Rightarrow -R D \sin(90 - \alpha) + C_y = 0 \\ \sum M_C = 0 &\Rightarrow M - R D \cos(90 - \alpha) (10 \sin 60^\circ) - R D \sin(90 - \alpha) (10 \cos 60^\circ) = 0 \end{aligned}$$

For entire system:

$$\begin{aligned} \sum M_A = 0 &\Rightarrow (M) + P \cdot y_A = 0 \\ \text{or} \\ \sum M_C = 0 &\Rightarrow (M) + y_A \cdot P = 0 \end{aligned}$$

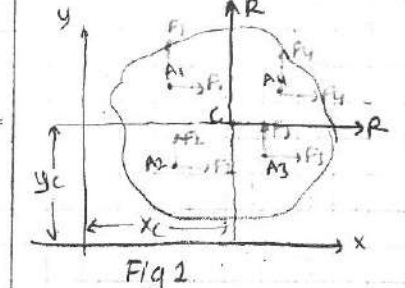
Center of Parallel Forces and Center of Gravity:

Let P and Q be two parallel forces applied to a rigid body at points A and B. Then the resultant R is parallel to the given forces, equal to their algebraic sum, and its line of action intersects the line AB at a point C such that (see Fig.)



Let $A_1, A_2, \dots, A_n$ be any system of points which all lie in one plane of a rigid body, and let $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$ be their coordinates. (See Fig 2). To find the center of parallel forces C for a given system of forces $F_1, F_2, \dots, F_n$ applied at these points and acting in any direction, let us first imagine that the forces act parallel to the y-axis. Then, we have a system of forces in one plane and the arm x_c of their resultant may be found by

38



$$\begin{aligned} x_c &= \frac{\sum (F_i x_i)}{\sum F_i} \\ y_c &= \frac{\sum (F_i y_i)}{\sum F_i} \end{aligned}$$

Center of gravity: The center of gravity of a body is that point through which the resultant of the distributed gravity force passes regardless of the body in space.

The term center of gravity will be used only with reference to actual physical bodies and the term centroid with reference to geometric figures and curves.

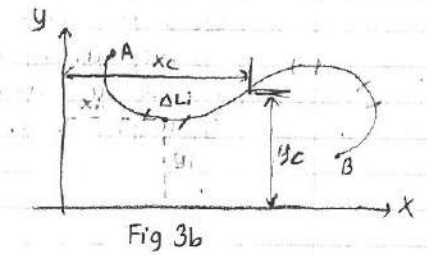
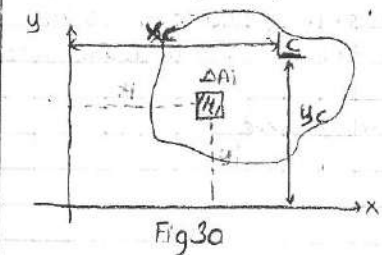
To define analytically the position of the centroid of area of a plane figure, we refer the area to coordinate axes (see Fig 3a), then denoting by ΔA_i the area of one element and by x_i, y_i the coordinates of its center, equations 1 and 2 will take the form

$$x_c = \frac{\sum (\Delta A_i x_i)}{\sum \Delta A_i} \quad (3); \quad y_c = \frac{\sum (\Delta A_i y_i)}{\sum \Delta A_i} \quad (4)$$

In the same manner, we may define the position of the centroid of length of a plane curve (see Fig 3b) by the formulas:

$$x_c = \frac{\sum (\Delta L_i x_i)}{\sum \Delta L_i} \quad (5); \quad y_c = \frac{\sum (\Delta L_i y_i)}{\sum \Delta L_i} \quad (6)$$

(ΔL_i denotes the length of an element and x_i, y_i the coordinates of its mid-point)

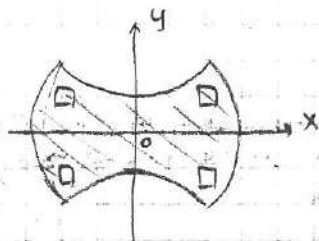


(39)

By making the elements ΔA_i or ΔL_i smaller and smaller, expressions (3)-(5)-(6) become more and more accurate. In the limit, when ΔA_i or ΔL_i become infinitesimal in size, expressions become exact. In such case, we write them in the form

$$\left\{ x_c = \frac{\int x dA}{\int dA}, y_c = \frac{\int y dA}{\int dA} \right\}; \left\{ x_c = \frac{\int x dL}{\int dL}, y_c = \frac{\int y dL}{\int dL} \right\}$$

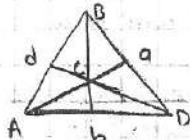
Remark: If a figure has two axes of symmetry, its centroid lies at their intersection.



$$(x_c = 0; y_c = 0)$$

17.11.16

$\Rightarrow$ The centroid of the area of any triangle lies at the intersection of its medians.



Guldinus

Theorems of Pappus: (or Pappus and ~~Stevinus~~)

Theorem 1: The area of a surface of revolution equals the product of the length of the generating curve and the distance traveled by the centroid of the curve in generating the surface area.

$$A(\text{surface Area}) = \theta \cdot \bar{r} \cdot L$$

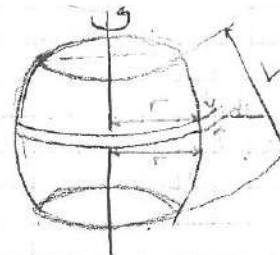
Where A = surface area of revolution.

θ = angle of revolution measured in radians, $\theta \leq 2\pi$.

$\bar{r}$ = perpendicular distance from the axis of revolution to the centroid of the generating curve.

L = length of the generating curve.

(40)



$$dA = 2\pi r dL$$

$$A = \int 2\pi r dL$$

$$= 2\pi \int r dL$$

$$= 2\pi \bar{r} L$$

$$\bar{r} = \frac{\int r dL}{\int dL} = \frac{\int r dL}{L}$$

$$\Rightarrow \int r dL = \bar{r} L$$

Theorem 2: The volume of a body of a revolution equals the product of the generating area and the distance traveled by the centroid of the area in generating the volume.

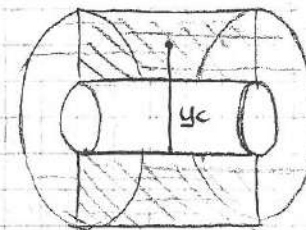
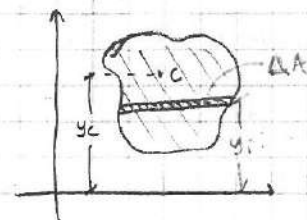
$$V(\text{volume}) = \theta \cdot \bar{r} \cdot A$$

Where V = volume of revolution

θ = angle of revolution measured in radians, $\theta \leq 2\pi$

$\bar{r}$ = perpendicular distance from the axis of revolution to the centroid of the generating area.

A = generating area.



$$V = 2\pi \bar{y} A$$

$$y_c = \frac{\int y dA}{\int dA} \Rightarrow \int y dA = y_c A$$

$$dV = 2\pi y dA$$

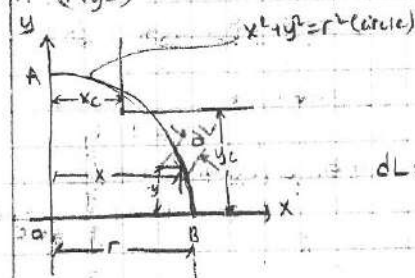
$$V = 2\pi \int y dA$$

$$V = 2\pi y_c A$$

Ex 5

(41)

Determine, by integration, the coordinates x_c and y_c of the centroid C of the quadrant AB of the arc of a circle of radius r . (Fig 1)



$$x_c = \frac{\int x dL}{\int dL}, \quad y_c = \frac{\int y dL}{\int dL}$$

$$dL = ds = \sqrt{1+y^2} dx = \sqrt{1+\frac{x^2}{r^2-x^2}} dx$$

$$= \frac{r}{\sqrt{r^2-x^2}} dx = \frac{r}{y} dx$$

Fig 1

$$y' = \frac{dy}{dx} = \frac{-2x}{2\sqrt{r^2-x^2}}$$

$$y = \sqrt{r^2-x^2}$$

$$dr^2 = dx^2 + dy^2$$

$$x_c = \frac{\int_0^r x \cdot \frac{r}{\sqrt{r^2-x^2}} dx}{\int_0^r \frac{r}{\sqrt{r^2-x^2}} dx}$$

$$r^2 - x^2 = u^2 \quad -2x dx = 2u du \Rightarrow r \int \frac{x dx}{\sqrt{r^2-x^2}} = r \int \frac{-u du}{u} = r \cdot (-u)$$

$$\Rightarrow r \cdot (-\sqrt{r^2-x^2}) \Big|_0^r = r^2$$

$$r \int \frac{dx}{\sqrt{r^2-x^2}} \Rightarrow r \int \frac{r \cos \theta d\theta}{r \cos \theta} = r \cdot \theta \Rightarrow r \cdot \text{Arcsin}\left(\frac{x}{r}\right) \Big|_0^r = r \left[\left(\frac{\pi}{2}\right) - 0 \right] = \frac{\pi r}{2}$$

$$x = r \sin \theta \quad dx = r \cos \theta d\theta$$

$$y = r \cos \theta$$

$$x_c = \frac{r^2}{\frac{\pi r}{2}} = \frac{2r}{\pi}$$

$$y_c = \frac{\int y dL}{\int dL} = \frac{\int_0^{\pi/2} y \cdot \frac{r}{y} dx}{\frac{\pi r}{2}} = \frac{2r}{\pi}$$

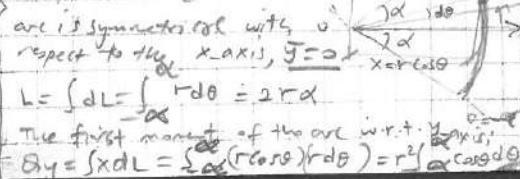
$$(x_c, y_c) = \left(\frac{2r}{\pi}, \frac{2r}{\pi} \right)$$

$$\bar{y} = \bar{x} \cdot L \Rightarrow \bar{x} \cdot (r\alpha) = 2r^2 \sin \alpha$$

$$\bar{x} = \frac{r \sin \alpha}{\alpha}$$

$$\bar{y} = 2r^2 \sin \alpha$$

Ex Determine the location of the centroid of the arc of a circle shown.



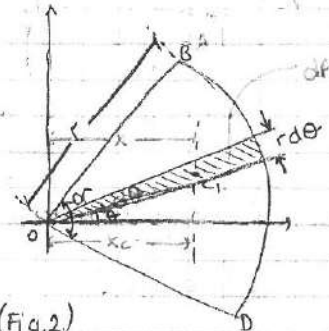
$$L = \int dL = \int_0^\alpha r d\theta = r\alpha$$

$$\bar{y} = \frac{\int y dL}{L} = \frac{\int_0^\alpha (r \cos \theta) (r d\theta)}{r\alpha} = \frac{r^2 \sin \alpha}{\alpha}$$

Ex 1

(42)

Determine the coordinates x_c and y_c of the centroid C of the area of the circular sector OBD of radius r and central angle α . (Fig 2)



(Fig 2)

$$dA = \frac{r(r d\theta)}{2} = \frac{r^2}{2} d\theta$$

$$x = \frac{2r}{3} \cos \theta$$

X-axis is symmetry axis:

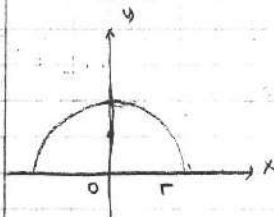
$$y_c = 0 \quad x_c = ?$$

$$x_c = \frac{\int x dA}{\int dA} = \frac{\int_0^{\alpha/2} \left(\frac{2r}{3} \cos \theta \right) \left(\frac{r^2}{2} d\theta \right)}{\int_0^{\alpha/2} \frac{r^2}{2} d\theta}$$

$$= \frac{\left(\frac{2r^3}{3} \right) \int_0^{\alpha/2} \cos \theta d\theta}{\left(\frac{r^2}{2} \right) \int_0^{\alpha/2} d\theta} = \frac{\frac{2r^3}{3} \sin \theta \Big|_0^{\alpha/2}}{r^2 \left(\frac{\alpha}{2} \right)}$$

$$x_c = \frac{4r}{3} \frac{\sin \alpha/2}{\alpha}$$

Ex:



$$x_c = 0, \quad y_c = \frac{4r}{3\pi}$$

$$\alpha = \pi \Rightarrow x_c = \frac{4r}{3\pi} \text{ (half disk)}$$

Centroids of composite plane figures and curves

We simply subdivide the figure into its several component parts, for which the centroids are already known and then treating each component part as an element.

$$x_c = \frac{A_1 x_1 + A_2 x_2 + \dots + A_n x_n}{A_1 + A_2 + \dots + A_n}$$

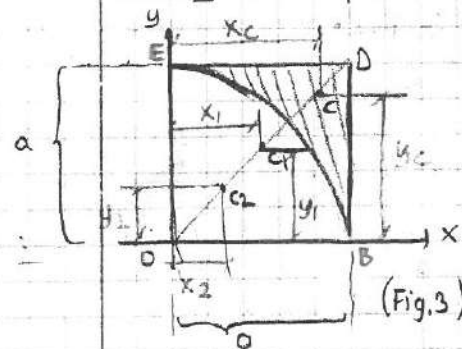
$$y_c = \frac{A_1 y_1 + A_2 y_2 + \dots + A_n y_n}{A_1 + A_2 + \dots + A_n}$$



Fig.

Ex: Locate the centroid C of the shaded area of the figure (Fig. 3) which is obtained by cutting the quadrant of a circle of radius a from a square OBDE of the same dimensions.

(43)

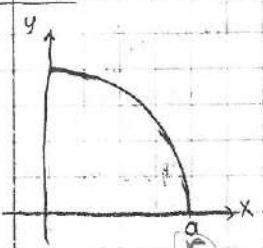


$$C_1: \begin{cases} (x_1, y_1) = \left(\frac{8a}{3\pi}, \frac{8a}{3\pi}\right) \\ A_1 = \frac{\pi a^2}{4} \end{cases}$$

$$C_2: \begin{cases} (x_2, y_2) = (a/2, a/2) \\ A_2 = a^2 \end{cases}$$

$$y_c = x_c = \frac{A_2 x_2 - A_1 x_1}{A_2 - A_1} = \frac{(a^2)(a/2) - \left(\frac{\pi a^2}{4}\right)\left(\frac{8a}{3\pi}\right)}{a^2 - \frac{\pi a^2}{4}}$$

22.11.2016



$$x_c = \frac{\int \tilde{x} dA}{\int dA} \quad \tilde{x} = \left(\frac{2r}{3}\right) \cos \theta$$

$$y_c = \frac{\int \tilde{y} dA}{\int dA} \quad \tilde{y} = \left(\frac{2r}{3}\right) \sin \theta$$

$$0 \leq \theta \leq \pi/2$$

$$x_c = y_c$$

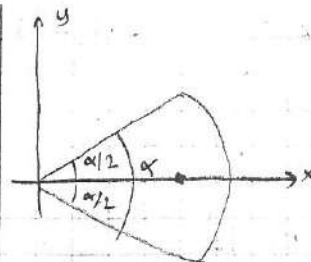
$$C: (x_c, y_c) = \left(\frac{4r}{3\pi}, \frac{4r}{3\pi}\right)$$

$$x_c = \frac{\int_0^{\pi/2} \left(\frac{2r}{3}\right) \cos \theta \frac{r^2 d\theta}{2}}{\int_0^{\pi/2} \frac{r^2 d\theta}{2}} = \frac{\left(\frac{r^3}{3}\right) \int_0^{\pi/2} \cos \theta d\theta}{\left(\frac{r^2}{2}\right) \int_0^{\pi/2} d\theta}$$

$$= \frac{\left(\frac{r^3}{3}\right)}{\frac{\pi r^2}{4}} = \frac{4r}{3\pi}$$

$$y_c = \frac{\int_0^{\pi/2} \left(\frac{2r}{3}\right) \sin \theta \left(\frac{r^2}{2} d\theta\right)}{\frac{\pi r^2}{4}} = \frac{\left(\frac{r^3}{3}\right) - \cos \theta \Big|_0^{\pi/2}}{\frac{\pi r^2}{4}} = \frac{4r}{3\pi}$$

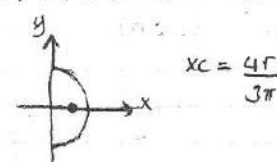
(44)



$$x_c = ?$$

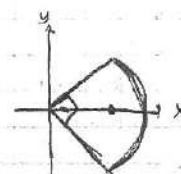
$$y_c = 0 \quad x_c = \frac{4r}{3\alpha} \sin \frac{\alpha}{2}$$

$$i) \alpha = \pi$$



$$x_c = \frac{4r}{3\pi}$$

$$ii) \alpha = \pi/2$$



$$x_c = \frac{8r}{3\pi} \left(\frac{\sin \frac{\pi}{4}}{4}\right) = \frac{4r \cdot \sqrt{2}}{3\pi}$$

Ex: A triangle ADE is to be cut from a square ABCD of dimension a as shown in Fig. 1. Find the altitude y of this triangle so that its vertex E will be the centroid of the remaining shaded area.

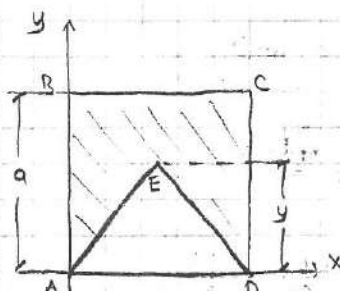
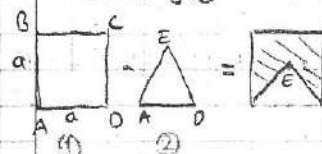


Fig. 1



$$x_c = \frac{\int x dA}{\int dA}, \quad y_c = \frac{\int y dA}{\int dA}$$

$$x_c = \frac{A_1 x_1 - A_2 x_2}{A_1 - A_2}; \quad y_c = \frac{A_1 y_1 - A_2 y_2}{A_1 - A_2}$$

$$① C_1(x_1, y_1) = \left(\frac{a}{2}, \frac{a}{2}\right)$$

$$A_1 = a^2$$

$$② C_2(x_2, y_2) = (x_2, y/3)$$

$$A_2 = \frac{a \cdot y}{2}$$

$$y_c = y$$

$$y_c = y = \frac{(a^2)(a/2) - \left(\frac{ay}{2}\right)(y/3)}{a^2 - \frac{ay}{2}}$$

$$\frac{a^3}{2} - \frac{ay^2}{6} = a^2 y - \frac{ay^2}{2}$$

$$3a^3 - ay^2 = 6a^2 y - 3ay^2$$

$$2ay^2 - 6a^2 y + 3a^3 = 0$$

$$\Delta = (-6a^2)^2 - (4)(2a)(3a^3)$$

$$= 36a^4 - 24a^4$$

$$= 12a^4$$

(45)

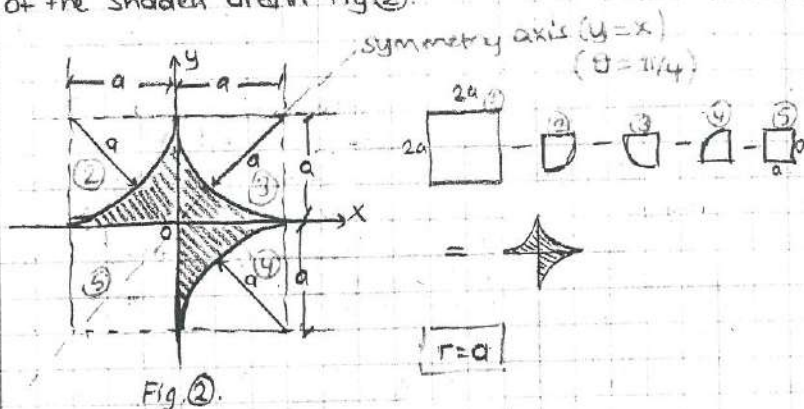
$$y_{1,2} = \frac{6a^2 \pm \sqrt{12a^4}}{2(2a)} = \frac{6a^2 \pm 2a^2\sqrt{3}}{4a} = \frac{3a \pm a\sqrt{3}}{2}$$

$$= a \left(\frac{3 \pm \sqrt{3}}{2} \right) \quad y_1 = \left(\frac{3 + \sqrt{3}}{2} \right) a = 2.309a \quad X$$

$$y_2 = \left(\frac{3 - \sqrt{3}}{2} \right) a = 0.6339a \quad \checkmark$$

EX:

With respect to the coordinate axes shown, locate the centroid of the shaded area in Fig. ②.



$$c_1(x_1, y_1) = (0, 0), \quad c_2(x_2, y_2) = \left(-a + \frac{4r}{3\pi}, \frac{a - 4r}{3\pi} \right),$$

$$A_1 = (2a)^2 = 4a^2, \quad A_2 = \frac{\pi a^2}{4}$$

$$c_3(x_3, y_3) = \left(a - \frac{4r}{3\pi}, a - \frac{4r}{3\pi} \right), \quad c_4(x_4, y_4) = \left(a - \frac{4r}{3\pi}, -a + \frac{4r}{3\pi} \right)$$

$$A_3 = \frac{\pi a^2}{4}, \quad A_4 = \frac{\pi a^2}{4}$$

$$c_5(x_5, y_5) = \left(-\frac{a}{2}, -\frac{a}{2} \right)$$

$$A_5 = a^2$$

$$x_c = \frac{A_1 x_1 - A_2 x_2 - A_3 x_3 - A_4 x_4 - A_5 x_5}{A_1 - A_2 - A_3 - A_4 - A_5}$$

$$y_c = \frac{A_1 y_1 - A_2 y_2 - A_3 y_3 - A_4 y_4 - A_5 y_5}{A_1 - A_2 - A_3 - A_4 - A_5}$$

(46)

$$= \frac{(4a^2)(0) - \left(\frac{\pi a^4}{4}\right)\left(-a + \frac{4a}{3\pi}\right) - \left(\frac{\pi a^4}{4}\right)\left(a - \frac{4a}{3\pi}\right) - \left(\frac{\pi a^4}{4}\right)\left(a - \frac{4a}{3\pi}\right) - (a^2)\left(-\frac{a}{2}\right)}{4a^2 - \left(\frac{\pi a^2}{4}\right)3 - a^2}$$

$$x_c = y_c = 0.0744 \cdot a$$

EX:

Determine the coordinates x_c and y_c of the centroid C of the area between the parabola $y = x^2/a$ and the straight line $y = x$ (Fig. 3).

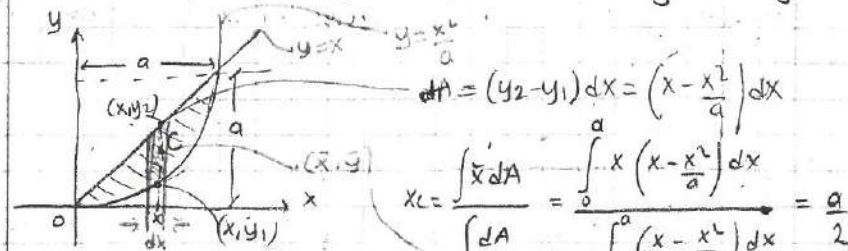


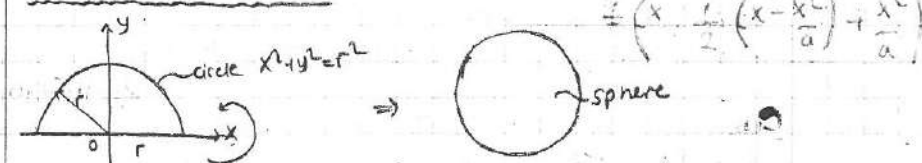
Fig. 3

$$dA = (y_2 - y_1) dx = \left(x - \frac{x^2}{a} \right) dx$$

$$x_c = \frac{\int x dA}{\int dA} = \frac{\int_0^a x \left(x - \frac{x^2}{a} \right) dx}{\int_0^a \left(x - \frac{x^2}{a} \right) dx} = \frac{a}{2}$$

$$y_c = \frac{\int y dA}{\int dA} = \frac{\int_0^a \left[\frac{1}{2} \left(x - \frac{x^2}{a} \right) + \frac{x^2}{a} \right] \left(x - \frac{x^2}{a} \right) dx}{\int_0^a \left(x - \frac{x^2}{a} \right) dx} = \frac{2a}{5}$$

Revolution about x -axis



Surface area of sphere: $4\pi r^2$

$$4\pi r^2 = 2\pi \cdot y_c \left(\frac{2\pi r}{2} \right)$$

$$V = 2\pi y_c A$$

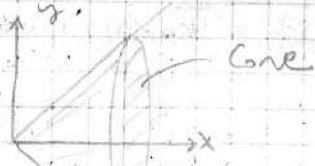
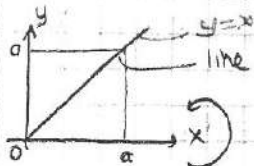
$$\frac{4\pi r^3}{3} = 2\pi y_c \left(\frac{\pi r^2}{2} \right)$$

$$S = 2\pi y_c \left(\frac{2\pi r}{4} \right)$$

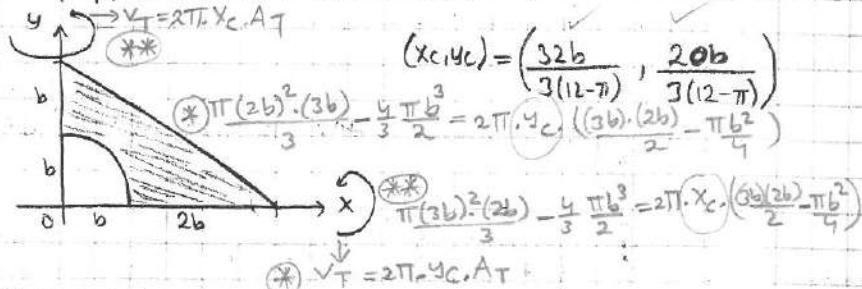
$$y_c = \frac{4r}{3\pi}$$

$$\frac{4\pi r^2}{2} = 2\pi y_c \left(\frac{2\pi r}{4} \right) \Rightarrow y_c = \frac{2r}{\pi}$$

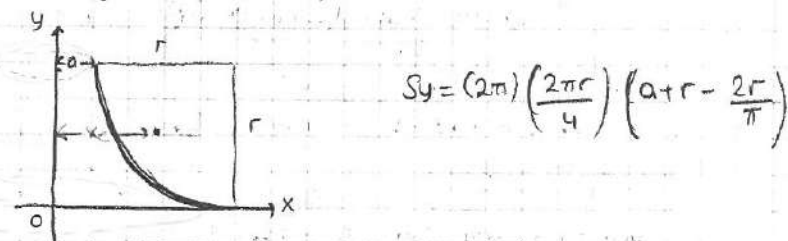
(47)



Ex: Find the centroid of the region (x_c, y_c) given below by using the pappus and Guldinus theorem:



Ex: Find the surface area of the arc given below by using pappus theorem (revolving about Oy -axis)



Cables:

24.11.2016

- 1) Cable subjected to Concentrated Loads
- 2) a) cable subjected to a Distributed Load
b) cable subjected to its Own Weight (Chain)

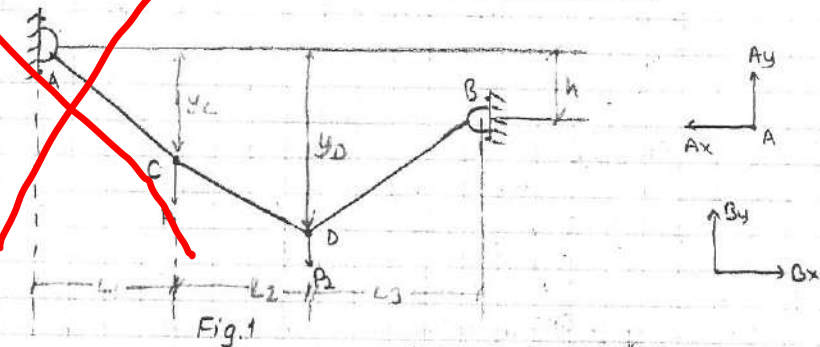
In engineering structures, we sometimes encounter flexible cables or chains suspended between two supports at their ends and subjected to the action of vertical load continuously distributed along their lengths. These structures are used for support and transmit loads from one member to another. We will make the assumption that the cable is perfectly flexible and inextensible.

(48)

That is, the tensile force acting in the cable is always tangent to the cable at points along its length.

Being inextensible, the cable has a constant length both before and after the load is applied.

1) Cable Subjected to Concentrated Loads

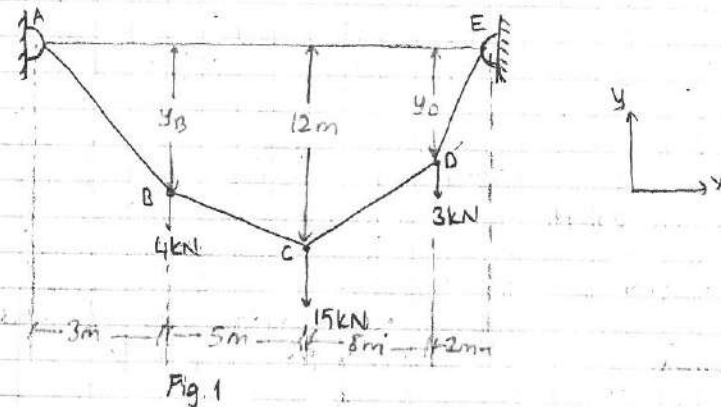


three equilibrium equations
 A_x, A_y, B_x, B_y four unknowns. $(\sum F_x = 0, \sum F_y = 0, \sum M_{A,B,C,D} = 0)$

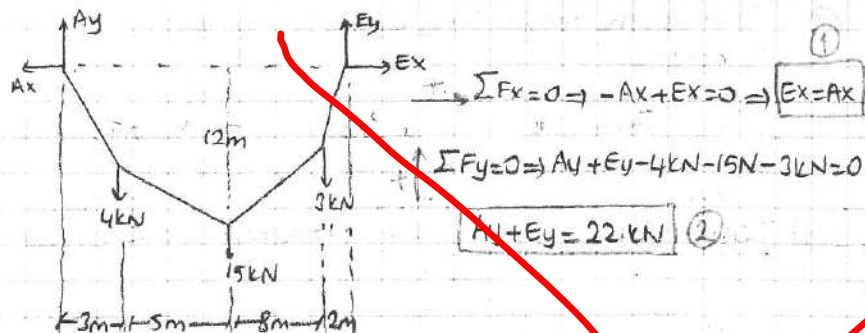
This type of problem can be solved by specifying one of the sags, either y_c or y_D , instead of the cable length.

By doing this, equilibrium equations are then sufficient for obtaining the unknown forces and remaining sag. (Fig. 1)

Ex: Determine the tension in each segment of the cable shown in Fig. 1.

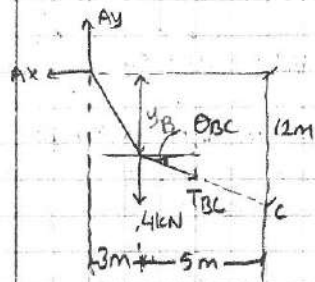


(49)



(3)

$$\sum M_E = 0 \Rightarrow -A_y(18\text{m}) + (4\text{kN})(15\text{m}) + (15\text{kN})(10\text{m}) + (3\text{kN})(2\text{m}) = 0$$



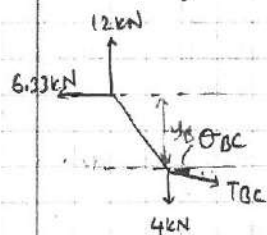
$$\sum M_C = 0 \Rightarrow A_x(12\text{m}) + (4\text{kN})(5\text{m}) - A_y(8\text{m}) = 0$$

$$(3) A_y = 12\text{kN}$$

$$(2) E_y = 10\text{kN}$$

$$(4) A_x = \frac{(12)(8) - 20}{12} = 8 - \frac{5}{3} = \frac{19}{3} = 6.33\text{kN}$$

$$(1) A_x = E_x = 6.33\text{kN}$$



$$\sum F_x = 0 \Rightarrow -6.33\text{kN} + T_{BC} \cos \theta_{BC} = 0$$

$$T_{BC} \cos \theta_{BC} = 6.33\text{kN}$$

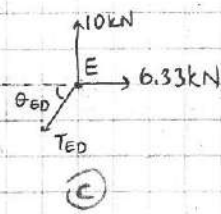
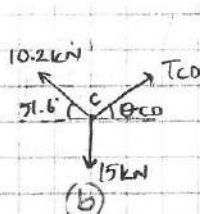
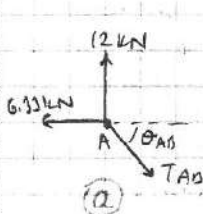
$$\sum F_y = 0 \Rightarrow 12\text{kN} - 4\text{kN} - T_{BC} \sin \theta_{BC} = 0$$

$$T_{BC} \sin \theta_{BC} = 8\text{kN}$$

$$\tan \theta_{BC} = \frac{8\text{kN}}{6.33\text{kN}} \Rightarrow \theta_{BC} = 51.6^\circ$$

$$T_{BC} = 10.2\text{kN}$$

Enforce cable
ABDE ACDE
or higher
equilibrium
knowledge
or y=10.2kN



(50)

$$\sum F_x = 0$$

$$-6.33\text{kN} + T_{AB} \cos \theta_{AB} = 0$$

$$\sum F_y = 0$$

$$12\text{kN} - T_{AB} \sin \theta_{AB} = 0$$

$$\theta_{AB} = 62.2^\circ$$

$$T_{AB} = 13.6\text{kN}$$

$$\tan \theta_{AB} = \frac{y_B}{3} \Rightarrow y_B = 3\text{m}$$

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\theta_{CD} = 47.9^\circ$$

$$T_{CD} = 9.44\text{kN}$$

$$\sum F_x = 0$$

$$\sum F_y = 0$$

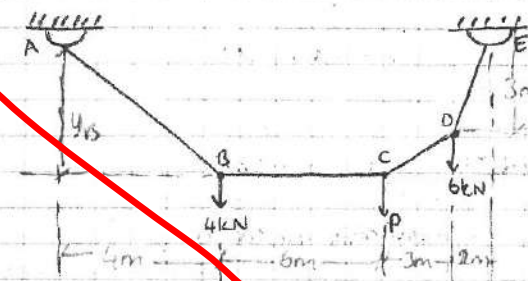
$$\theta_{ED} = 57.7^\circ$$

$$T_{ED} = 11.8\text{kN}$$

$$\tan \theta_{ED} = \frac{y_D}{2} \Rightarrow y_D = 2\text{m}$$

Remark: By comparison, the maximum cable tension is in segment AB since this segment has the greatest slope (θ) and its required that for any cable segment the horizontal component $T \cos \theta = A_x = E_x$.

HW: Determine the force P needed to hold the cable in the position shown, i.e., so segment BC remains horizontal. Also, compute the sag y_B and the maximum tension in the cable.



Answer

$$P = 0.8\text{kN}$$

$$T_{AB} = 6.05\text{kN}$$

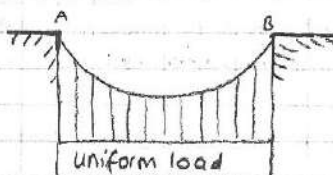
$$T_{BC} = 4.53\text{kN}$$

$$T_{CD} = 4.60\text{kN}$$

$$T_{max} = T_{DE} = 8.17\text{kN}$$

$$y_B = 3.53\text{m}$$

2) a) Cables Subjected to Distributed Loading



(a) a thin wire cable or cord.

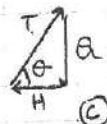
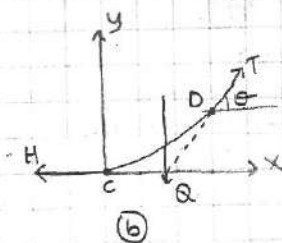
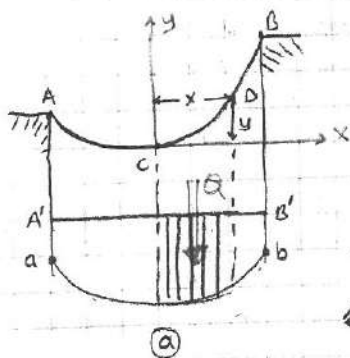
in discussing such cables, two questions at once present

① What is the slope of the curve that the chain or cord will assume in the condition of equilibrium?

② how does the tension vary from point to point along the curve?

(51)

каблон
длина l
генератор T
загрузка q
нагрузка H (горизонтальная)
каблон



Q : the vertical force representing that the portion of the distributed load between C and D .

H, T : two tensile forces representing the reactions exerted by the unconsidered portions of the chain on either side. The forces H and T are tangent to the curve at C and D , respectively.

From figure c: $\tan \theta = \frac{Q}{H}$

$$\text{(slope)} \leftarrow \frac{dy}{dx} = \frac{Q}{H} \quad (1)$$

$$T_{\max} = \sqrt{H^2 + Q^2} \quad (2)$$

Eq (1) is the differential equation of the curve of equilibrium assumed by the cable under the action of the load that it carries, while eq.(2) defines the tension at any point on the curve

$$Q = q \cdot x$$

Parabolic cable: Assuming that the cable in Fig.(a,b,c) carries vertical load of intensity q uniformly distributed with respect to the horizontal span l , Eq.1 becomes

$$\frac{dy}{dx} = \frac{qx}{H} \quad (3)$$

(52)

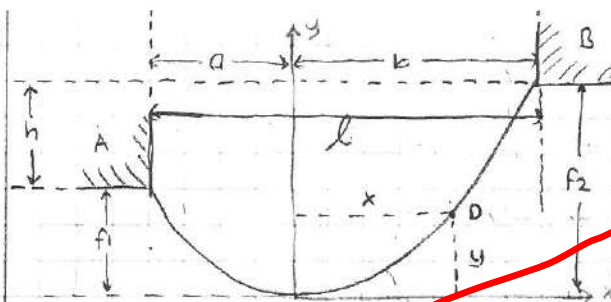


Fig. 2

$\frac{q}{H}$ is constant, integrate once we obtain

$$\int dy = \int \frac{qx}{H} dx \Rightarrow y = \frac{q}{H} \frac{x^2}{2} + C \quad x=0, y=0 \Rightarrow C=0 \quad (\text{integration constant})$$

$$y = \frac{q}{2H} x^2 \quad (4)$$

Eq (4) is the equation of the curve of equilibrium assumed by the cable and it is a portion of a parabola with vertical axis.

From eq.(2)

$$T_{\max} = \sqrt{H^2 + (qx)^2} \quad (5)$$

from which we note that the tension in the cable is a minimum at the lowest point C , where it is equal to H , and it increases toward the ends of the cable, being a maximum at the higher support. To determine the distances a and b locating the low point C with reference to the supports A and B , we use eq.(4) successively for the portions AC and BC of the cable and note that when $x=-a$, $y=f_1$ and $x=b$, $y=f_2$

$$x=-a, y=f_1 \Rightarrow f_1 = \frac{q(-a)^2}{2H} = \frac{qa^2}{2H}$$

$$x=b, y=f_2 \Rightarrow f_2 = \frac{q(b)^2}{2H} = \frac{qb^2}{2H} \quad (7)$$

$$\text{from Eq(5): } \begin{cases} T_a = \sqrt{H^2 + (qa)^2} \\ T_b = \sqrt{H^2 + (qb)^2} \end{cases} \quad (6)$$

(53)

Substituting the first of these equations (Eq (7)) from the second and using the notation $f_2 - f_1 = h$, we obtain $h = \frac{q}{2H} (b^2 - a^2)$

$$2hH = q(b^2 - a^2), \quad a + b = l$$

$$2hH = q((l-a)^2 - a^2) \Rightarrow 2hH = q(l^2 - 2al + a^2 - a^2)$$

$$\Rightarrow \frac{2hH}{q} = l - 2a \Rightarrow \begin{cases} a = \frac{l}{2} - \frac{hH}{q} \\ b = \frac{l}{2} + \frac{hH}{q} \end{cases} \quad (8)$$

when $h=0$, that is, when the supports A and B are on the same level, we obtain $a=b=l/2$. Substituting the above expression for b into the second of eqs(7), the following equation for calculating H is obtained:

$$F_2 = \frac{q \left(\frac{l}{2} + \frac{hH}{q} \right)^2}{2H} \Rightarrow \frac{2H}{q} \cdot f_2 = \frac{l^2}{4} + \frac{hH}{q} + \frac{h^2}{q^2} \cdot \frac{H^2}{l^2}$$

$$\Rightarrow H^2 - \frac{2ql^2}{h^2} (f_2 - h) \cdot H + \frac{q^2 l^4}{4h^2} = 0 \quad \left(H_{1,2} = \frac{-b \pm \sqrt{\Delta}}{2a} \right)$$

$$\text{from which } H = \frac{ql^2}{h^2} \left(f_2 - \frac{h}{2} \pm \sqrt{f_1 f_2} \right) \quad (9)$$

In eq.(9), the minus sign should be used for all cases where the vertex of the parabola corresponding to the configuration of equilibrium of the cable lies between the supports as shown in Fig. 2, while the plus sign should be used for all cases where the vertex of their parabola lies to the same side of both supports.

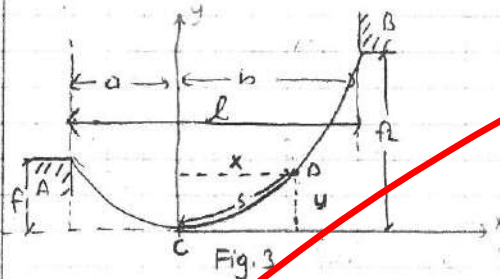
In the particular case, where the supports are on the same level, $f_1 = f_2 = f$, $a = b = l/2$

$$f = \frac{q(l/2)^2}{2H} \Rightarrow \boxed{H = \frac{ql^2}{8f}} \quad (10)$$

from Eq.(9) and (10) the minimum tension H in the cable may be found.

(54) Catenary Cable: Assuming that the cable in Fig. 3 hangs freely in the gravity field and is subjected only to its own weight uniformly distributed along the curve, Eq(1) becomes

$$\frac{dy}{dx} = \frac{q \cdot s}{H} \quad (11) \quad \begin{array}{l} q \text{ is the weight per unit length of the cable and,} \\ s \text{ is the length of the arc CD} \end{array}$$



It will be necessary to express the length s as a function of x and y :

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + (y')^2} dx$$

$$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx} \right)^2} \Rightarrow \frac{ds}{dx} = \sqrt{1 + \left(\frac{qs}{H} \right)^2} \Rightarrow \frac{ds}{\sqrt{1 + \left(\frac{qs}{H} \right)^2}} = dx$$

$$\Rightarrow \int \frac{ds}{\sqrt{1 + \left(\frac{qs}{H} \right)^2}} = \int dx = x + C_1$$

$$\frac{qs}{H} = \sinh z \Rightarrow s = \frac{H}{q} \sinh z$$

$$ds = \frac{H}{q} \cosh z dz$$

$$x + C_1 = \frac{H}{q} \operatorname{Arsinh} \left(\frac{qs}{H} \right) \Rightarrow x = \frac{H}{q} \operatorname{Arsinh} \left(\frac{qs}{H} \right) \Rightarrow \frac{qx}{H} = \operatorname{Arsinh} \left(\frac{qs}{H} \right)$$

$$x=0 \quad s=0 \Rightarrow C_1=0$$

$$\Rightarrow \frac{qs}{H} = \sinh \left(\frac{qx}{H} \right)$$

$$\boxed{s = \frac{H}{q} \sinh \left(\frac{qx}{H} \right)} \quad (12)$$

$$\sinh z = \frac{e^z - e^{-z}}{2}$$

$$\cosh z = \frac{e^z + e^{-z}}{2}$$

$$\cosh^2 z - \sinh^2 z = 1$$

$$\frac{d}{dz} (\sinh z) = \cosh z$$

$$\frac{d}{dz} (\cosh z) = \sinh z$$

(55)

Substituting the s in Eq(11):

$$\frac{dy}{dx} = \frac{q}{H} \left(\frac{H}{q} \sinh\left(\frac{qx}{H}\right) \right) = \sinh\left(\frac{qx}{H}\right)$$

$$\Rightarrow \boxed{dy = \sinh\left(\frac{qx}{H}\right) dx} \quad (13)$$

Integrate once, we obtain $y = \frac{H}{q} \cosh\left(\frac{qx}{H}\right) + C_2$

$$x=0, y=0 \Rightarrow 0 = \frac{H}{q} + C_2 \Rightarrow C_2 = -\frac{H}{q}$$

$$y = \frac{H}{q} \left(\cosh\left(\frac{qx}{H}\right) - 1 \right) \quad (14)$$

The equation of the curve of equilibrium assumed by the cable when hanging freely under its own weight and represents a catenary with vertical axis.

$$Q = qs$$

$$Q = q \left(\frac{H}{q} \sinh\left(\frac{qx}{H}\right) \right) = H \sinh\left(\frac{qx}{H}\right)$$

$$T_a = H + qf_1$$

$$T_b = H + qf_2$$

$$T_{\max} = \sqrt{H^2 + Q^2} = H \cosh\left(\frac{qx}{H}\right) \Rightarrow \boxed{T_{\max} = H + qy} \quad (15)$$

$$y=0 \Rightarrow \boxed{T_{\min} = H}$$

$$x=-a, y=f_1 \Rightarrow f_1 = \frac{H}{q} \left(\cosh\left(\frac{qa}{H}\right) - 1 \right)$$

$$x=b, y=f_2 \Rightarrow f_2 = \frac{H}{q} \left(\cosh\left(\frac{qb}{H}\right) - 1 \right)$$

$$a = \frac{H}{q} \operatorname{arccosh}\left(\frac{qf_1}{H} + 1\right)$$

$$a+b=l$$

$$b = \frac{H}{q} \operatorname{arccosh}\left(\frac{qf_2}{H} + 1\right)$$

(56)

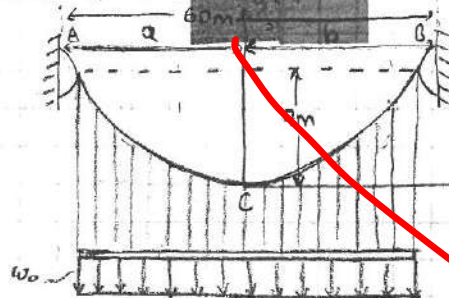
$$\frac{ql}{H} = \operatorname{arccosh}\left(\frac{qf_1}{H} + 1\right) + \operatorname{arccosh}\left(\frac{qf_2}{H} + 1\right)$$

for H , solved numerically

(57)

01.12.2016

Determine the maximum uniform distributed loading w_0 N/m that the cable can support if it is capable of sustaining a maximum tension of 60 kN.



$$\frac{dy}{dx} = \frac{Q}{H} \Rightarrow \frac{dy}{dx} = \frac{qx}{H} = \frac{w_0 x}{H}$$

$$x dy = \frac{w_0 x^2}{H} \Rightarrow y = \frac{w_0 x^2}{2H} + C_1$$

$$Q = \int q(x) dx \quad x=0, y=0 \Rightarrow C_1=0$$

$$y = \frac{w_0 x^2}{2H} \quad \text{the curve of the cable.}$$

$$x=30, y=7m \Rightarrow H = \frac{w_0 x^2}{2y}$$

$$H = \frac{w_0 (30)^2}{2(7)} = \frac{450 w_0}{7} \quad \text{the minimum cable tension.}$$

$$\begin{aligned} f_2 = f_1 = f = 7m \\ a = b = l/2 = 30m \\ T_{max} = 60kN \end{aligned}$$

$$dQ = d \int q(x) dx$$

$$= q(x)$$

$$\left(\frac{d^2 y}{dx^2} = \frac{q(x)}{H} = \frac{w_0}{H} \right)$$

$$q(x) = w_0$$

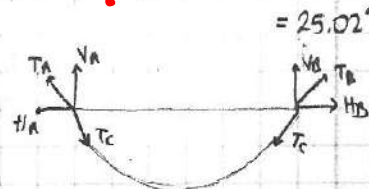
$$T_{max} = \sqrt{H^2 + (w_0 x)^2} \Rightarrow x=30m \Rightarrow 60 = \sqrt{\left(\frac{450 w_0}{7} \right)^2 + (30 w_0)^2} = w_0 \sqrt{\left(\frac{450}{7} \right)^2 + 900}$$

$$\Rightarrow w_0 = \frac{60kN}{\sqrt{\left(\frac{450}{7} \right)^2 + 900}} = 0.846 \text{ kN/m}$$

$$\text{2nd solution } T_{max} \cos \theta_{max} = H$$

$$\frac{dy}{dx} \Big|_{x=30} = \tan \theta_{max} = \frac{w_0 x}{H} = \frac{w_0 (30m)}{\frac{450 w_0}{7}} \Rightarrow \tan \theta_{max} = \frac{7}{15} \Rightarrow \theta_{max} = \arctan\left(\frac{7}{15}\right)$$

$$\frac{60 \cos(25.02^\circ)}{450/7} = w_0 = 0.846 \text{ kN/m.}$$



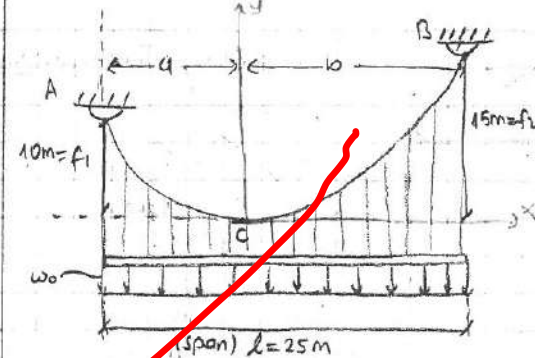
(58)

$$T_A \sin \theta_A + T_B \sin \theta_B = Q; H = H_A = H_B; y = \frac{w_0 x^2}{2H}$$

$$f = \frac{w_0 l^2}{8H} \Rightarrow H = \frac{w_0 l^2}{8f}$$

Ex:

The cable supports the uniform distributed load of $w_0 = 600 \text{ N/m}$. Determine the tension in the cable at each support A and B.



$$q = w_0 = 600 \text{ N/m}$$

$$y = \frac{qx^2}{2H} \Rightarrow y = \frac{(600 \text{ N/m}) x^2}{2H}$$

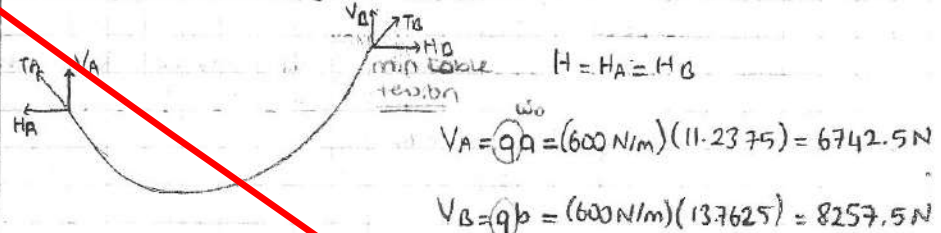
$$f_2 = \frac{qb^2}{2H} \Rightarrow 15 = \frac{600 b^2}{2H} \quad \left. \begin{aligned} b^2 &= 1.5 \\ a^2 &= 1.5 \end{aligned} \right\}$$

$$f_1 = \frac{qa^2}{2H} \Rightarrow 10 = \frac{600 a^2}{2H}$$

$$a + b = l = 25$$

$$a = 25/2.2247 = 11.2375 \quad b = 25 - 11.2375 = 13.7625$$

$$f_2 = \frac{w_0 b^2}{2H} \Rightarrow H = \frac{w_0 b^2}{2f_2} = \frac{600 (13.7625)^2}{2(15)} = 3788.13 \text{ N}$$



$$H = H_A = H_B$$

$$V_A = Qa = (600 \text{ N/m})(11.2375) = 6742.5 \text{ N}$$

$$V_B = Qb = (600 \text{ N/m})(13.7625) = 8257.5 \text{ N}$$

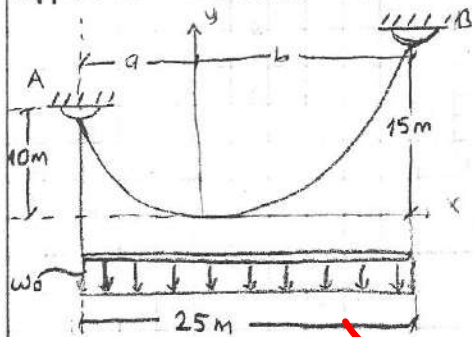
$$T_B = \sqrt{H_B^2 + V_B^2} = \sqrt{3788.13^2 + 8257.5^2} = 9083.58 \text{ N}$$

$$T_A = \sqrt{H_A^2 + V_A^2} = \sqrt{3788.13^2 + 6742.5^2} = 7735.08 \text{ N}$$

$$T_{max} = T_B \rightarrow T_{max} = \frac{H}{\cos \theta_{max}} \quad x = b$$

Ex:

Determine the maximum uniform distributed load w_0 the cable can support if the maximum tension in the cable can sustain is 4000 N.



$$a = 11.2375, b = 13.7625 \text{ m}$$

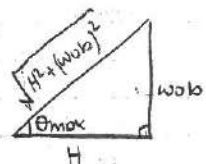
$$f_2 = \frac{w_0 b^2}{2H} \Rightarrow H = \frac{w_0 (13.7625)^2}{2f_2}$$

$$T_{\max} = 4000 \text{ N}; T_{\max} \cos \theta_{\max} = H$$

$$\frac{dy}{dx} = \frac{w_0 x}{H} = \tan \theta_{\max}$$

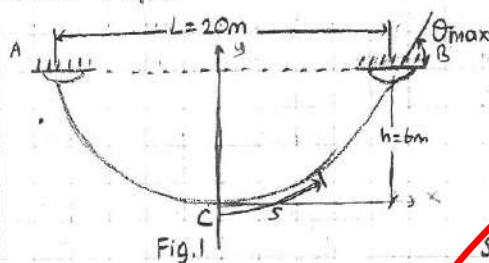
$$\cos \theta_{\max} = \frac{H}{\sqrt{H^2 + (w_0 b)^2}}$$

$$4000 \cdot \frac{H}{\sqrt{H^2 + (w_0 b)^2}} = H \Rightarrow w_0 = 264 \text{ N/m}, \theta_{\max} = 65.36^\circ$$



$$\theta_{\max} = \arctan\left(\frac{w_0 (13.7625)}{H}\right)$$

Determine the deflection curve, the length, and the maximum tension in the uniform cable shown in Fig.1. The cable has a weight per unit length of $w_0 = 5 \text{ N/m}$.



$$\frac{dy}{dx} = \frac{Q}{H} = \frac{w_0 s}{H} \Rightarrow ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$s = \int \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\frac{dy}{dx} = \frac{q \cdot s}{H}$$

$$\sinh\left(\frac{qx}{H}\right) \Rightarrow y = \frac{H}{q} \cosh\left(\frac{qx}{H}\right) + C \Rightarrow C = -\frac{H}{q}$$

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$$y = \frac{H}{q} \left(\cosh\left(\frac{qx}{H}\right) - 1 \right) \text{ m (deflection curve)} \Rightarrow H?$$

$$x = 10 \text{ m}, y = h = 6 \text{ m} \Rightarrow 6 = \frac{H}{5} \left(\cosh\left(\frac{5 \cdot 10}{H}\right) - 1 \right)$$

$$1 + \frac{30}{H} = \cosh\left(\frac{50}{H}\right) \Rightarrow H = 45.9 \text{ N}$$

$$\text{length} = \frac{s}{2} = \frac{H}{q} \sinh\left(\frac{qx}{H}\right) \Rightarrow \frac{s}{2} = \frac{45.9}{5} \left(\sinh\left(\frac{50}{45.9}\right) \right)$$

$$x = 10 \text{ m}$$

$$s = 24.2 \text{ m}$$

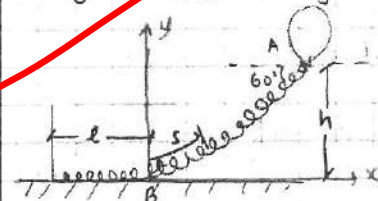
$$T_{\max} \cos \theta_{\max} = H$$

$$T_{\max} = \frac{45.9}{\cos 52.8^\circ} = 75.9 \text{ N}$$

$$\frac{dy}{dx} = \frac{5 (12.1)}{45.9} = \tan \theta_{\max} \Rightarrow \theta_{\max} = 52.8^\circ$$

HW:

The balloon is held in place using a 400 m cord that weighs 0.3 N/m and makes a 60° angle with the horizontal. If the tension in the cord at point A is 150 N, determine the length of the cord, ℓ , that is lying on the ground and the height.

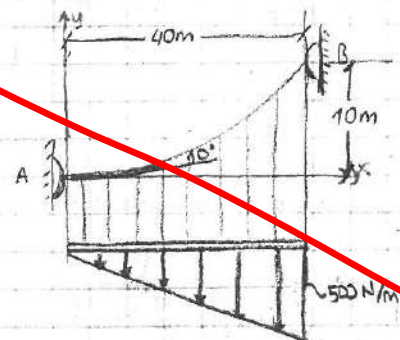


$$\ell = 238 \text{ m}$$

$$h = 93.75 \text{ m}$$

Ex:

If the slope of the cable at support A is 10° , determine the deflection curve $y = f(x)$ of the cable and the maximum tension developed in the cable.



$$\frac{dy}{dx} = \frac{Q}{H} \quad Q = \int q(x) dx$$

$$\frac{d^2 y}{dx^2} = \frac{q(x)}{H}$$

$$q(x) = ax + b$$

$$x = 0, y = 0 \Rightarrow q(0) = 0 = b$$

$$x = 40, y = 500 \Rightarrow 500 = a \cdot 40$$

$$a = \frac{500}{40}$$

$$q(x) = \frac{500}{40} x$$

mesnet
benzet
farkli
Ayagi
yuzey
belirli
anelli
dijgin
yagili
bir form
olmog
tam 2-
mer kebe
foran
kullan
q(x)
dijgu
dijgu
yuzey

1

$$\frac{d^2y}{dx^2} = \frac{500x}{40} = \frac{50}{4}x$$

$$\frac{dy}{dx} = \frac{50}{8}x^2 + C_1$$

$$x=0 \quad \frac{dy}{dx} = \tan 10^\circ \Rightarrow C_1 = \tan 10^\circ \Rightarrow \frac{dy}{dx} = \frac{50x^2}{8} + \tan 10^\circ$$

$$y = (46 \cdot 10^{-6})x^3 + 0.176x$$

$$T_{max} = 48.69 \times 10^3 \text{ N}$$

$$= 48685.85555 \text{ N}$$

Force systems in space:

Equilibrium in three dimensions:

$$\vec{F} = F_x \vec{i} + F_y \vec{j} + F_z \vec{k}$$

$$\vec{R} = \vec{F} = \sum_{i=1}^n (F_x)_i \vec{i} + \sum_{i=1}^n (F_y)_i \vec{j} + \sum_{i=1}^n (F_z)_i \vec{k}$$

$$\sum (F_x)_i = 0, \sum (F_y)_i = 0, \sum (F_z)_i = 0; \quad \vec{R} = 0$$

$$\vec{M}_i = M_{ix} \vec{i} + M_{iy} \vec{j} + M_{iz} \vec{k}$$

$$\vec{M} = \sum_{i=1}^n (M_x)_i \vec{i} + \sum_{i=1}^n (M_y)_i \vec{j} + \sum_{i=1}^n (M_z)_i \vec{k}$$

$$\sum (M_x)_i = 0, \sum (M_y)_i = 0, \sum (M_z)_i = 0; \quad \vec{M} = 0$$

$$\vec{F} = (F_1)\vec{i} + (F_2)\vec{j} + (F_3)\vec{k}$$

Scalar components

$$\vec{F} \cdot \vec{i} = F_1 \quad \vec{F} \cdot \vec{j} = F_2 \quad \vec{F} \cdot \vec{k} = F_3$$

$$\vec{F} \cdot \vec{j} = F_2$$

$$\vec{F} \cdot \vec{k} = F_3$$

$$|\vec{F}| = F$$

Scalar

$$R = \sqrt{\left(\sum_{i=1}^n (F_x)_i\right)^2 + \left(\sum_{i=1}^n (F_y)_i\right)^2 + \left(\sum_{i=1}^n (F_z)_i\right)^2}, \quad R = |\vec{R}|$$

$$M = \sqrt{\left(\sum_{i=1}^n (M_x)_i\right)^2 + \left(\sum_{i=1}^n (M_y)_i\right)^2 + \left(\sum_{i=1}^n (M_z)_i\right)^2}, \quad M = |\vec{M}|$$

$$x = 40 \text{ m} \quad y = 10 \text{ m} \Rightarrow$$

$$H = 45244.96709 \text{ N}$$

$$\Rightarrow y = \frac{50x^3}{24H} + (\tan 10^\circ)x + C_2$$

$$x=0, y=0 \Rightarrow C_2 = 0$$

$$\frac{dy}{dx} = \tan \theta_{max} = \frac{50(40)^2}{8(45244.96709)} + \tan 10^\circ$$

$$\tan \theta_{max} = 0.397356039$$

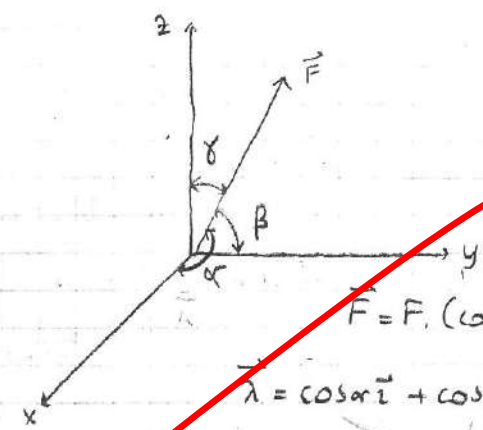
$$\theta_{max} = \arctan(\dots)$$

$$\theta_{max} = 21.67020275^\circ$$

$$T_{max} = \frac{H}{\cos(21.67020275^\circ)}$$

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$$F_x = F_1 = F \cos \alpha$$

$$F_y = F_2 = F \cos \beta$$

$$F_z = F_3 = F \cos \gamma$$

$$\vec{F} = F (\cos \alpha \vec{i} + \cos \beta \vec{j} + \cos \gamma \vec{k})$$

$$\vec{\lambda} = \cos \alpha \vec{i} + \cos \beta \vec{j} + \cos \gamma \vec{k}$$

directional cosines

Scalar Form of the Moment:

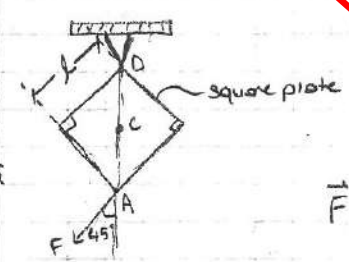
$$M = F \cdot d$$

Vector Form of the Moment:

$$\vec{M} = \vec{r} \times \vec{F}$$

Ex:

A 2m square plate hangs from one of its corners as shown in Fig. 1. At the diagonally opposite end, a force of 50 N is applied by pulling on the string AB. Find the moment of the applied force about the center C of the plate.



$$\sqrt{l^2 + l^2} = |DA| = \sqrt{2}l \quad ; |CA| = \frac{l}{\sqrt{2}}$$

$$\vec{r}_{CA} = -\frac{l}{\sqrt{2}} \vec{j}$$

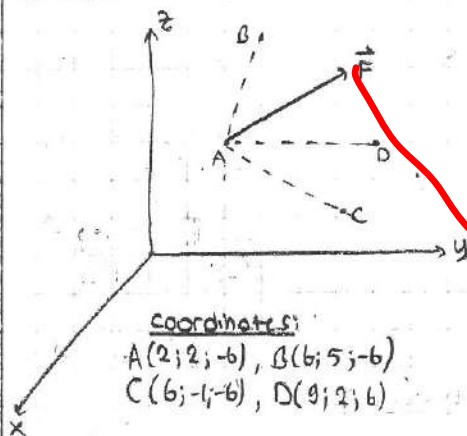
$$\vec{F} = |\vec{F}| (-\sin \theta \vec{i} - \cos \theta \vec{j}) = \frac{50 \text{ N}}{\sqrt{2}} \left(-\frac{1}{\sqrt{2}} \vec{i} - \frac{1}{\sqrt{2}} \vec{j} \right)$$

$$\vec{M}_C = \left(-\frac{l}{\sqrt{2}} \right) \vec{j} \times \left(-\frac{50}{\sqrt{2}} \vec{i} - \frac{50}{\sqrt{2}} \vec{j} \right) = -\frac{50}{2} l \vec{k} = -50 \text{ Nm} \vec{k}$$

$$l = 2 \text{ m}$$

(63)

Ex: Resolve the force $\vec{F}$ into its components in the AB, AC and AD directions:



$$\vec{F} = 50\vec{i} + 20\vec{j} + 100\vec{k} \text{ N}$$

$$\vec{F} = T_{AB}\vec{\lambda}_{AB} + T_{AD}\vec{\lambda}_{AD} + T_{AC}\vec{\lambda}_{AC} = 50\vec{i} + 20\vec{j} + 100\vec{k} \text{ N}$$

Scalars

$$T_{AB} \left(\frac{4\vec{i} + 3\vec{j}}{5} \right) + T_{AD} \left(\frac{7\vec{i} + 12\vec{j}}{\sqrt{49+144}} \right) + T_{AC} \left(\frac{4\vec{i} - 3\vec{j}}{5} \right) = 50\vec{i} + 20\vec{j} + 100\vec{k}$$

$$\frac{4}{5} T_{AB} + \frac{7}{14} T_{AD} + \frac{4}{5} T_{AC} = 50$$

$$\frac{3}{5} T_{AD} - \frac{3}{5} T_{AC} = 20$$

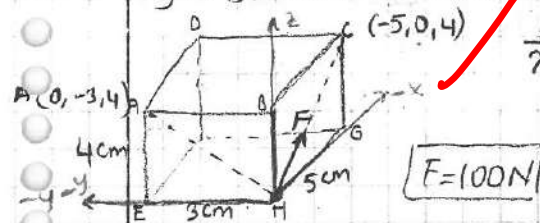
$$\frac{12}{14} T_{AD} = 100 \rightarrow T_{AD} = 115.75 \text{ N}$$

$$\begin{cases} T_{AB} = 11.46 \text{ N} \\ T_{AC} = -21.875 \text{ N} \\ T_{AD} = 115.75 \text{ N} \end{cases}$$

$T_{AD}, T_{AC} \checkmark$

Ex:

A rectangular prism is acted upon by a force $\vec{F}$ as shown in Fig 2. Determine the moment of $\vec{F}$ about A.



$$\vec{\lambda}_{AG} = \frac{-5\vec{i} + 0\vec{j} + 4\vec{k}}{\sqrt{(-5)^2 + (4)^2}} = \frac{-5}{\sqrt{41}}\vec{i} + \frac{4}{\sqrt{41}}\vec{k}$$

(64)

$$\vec{F} = F\vec{\lambda}_{HC} = \frac{-500}{\sqrt{41}}\vec{i} + \frac{400}{\sqrt{41}}\vec{k}$$

$$\vec{\lambda}_{AH} = \frac{3\vec{j} - 4\vec{k}}{\sqrt{(3)^2 + (-4)^2}} = \frac{3\vec{j} - 4\vec{k}}{5} \Rightarrow \vec{M}_A = \vec{\lambda}_{AH} \times \vec{F}$$

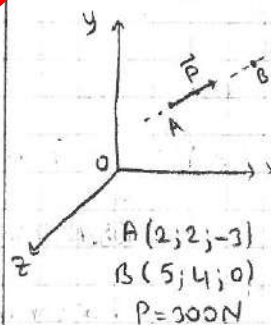
$$\vec{r}_{AH} = 3\vec{j} - 4\vec{k}$$

$$\vec{M}_A = \frac{1200}{5\sqrt{41}}\vec{i} + \frac{400}{\sqrt{41}}\vec{j} + \frac{900}{\sqrt{41}}\vec{k}$$

$$\begin{aligned} &= \left(\frac{3\vec{j} - 4\vec{k}}{5} \right) \times \left(\frac{-500\vec{i}}{\sqrt{41}} + \frac{400\vec{k}}{\sqrt{41}} \right) \\ &= \frac{900}{\sqrt{41}}\vec{i} + \frac{1200}{5\sqrt{41}}\vec{j} + \frac{400}{\sqrt{41}}\vec{k} \end{aligned}$$

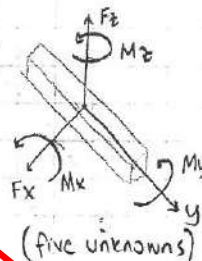
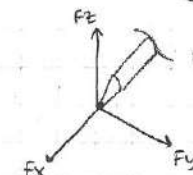
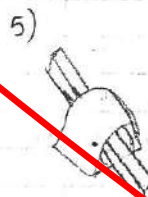
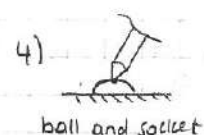
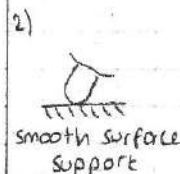
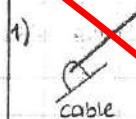
Ex:

Resolve the given force $\vec{P}$ into a force at O (origin) and a couple.



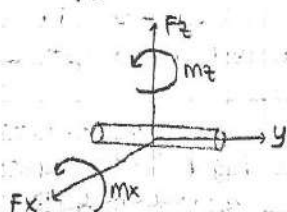
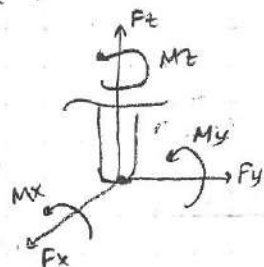
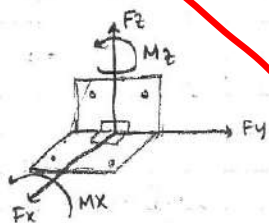
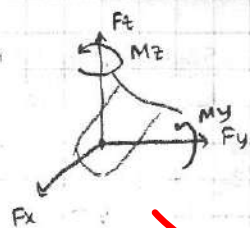
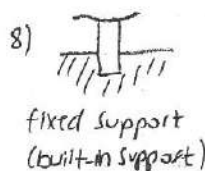
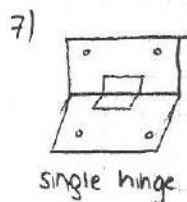
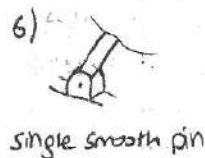
$$M = F \cdot d$$

Supports for Rigid Bodies Subjected to three-dimensional force systems



Suriname ydwa tepu ydwa dila!

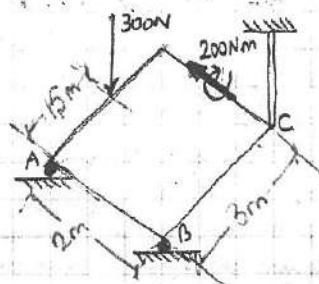
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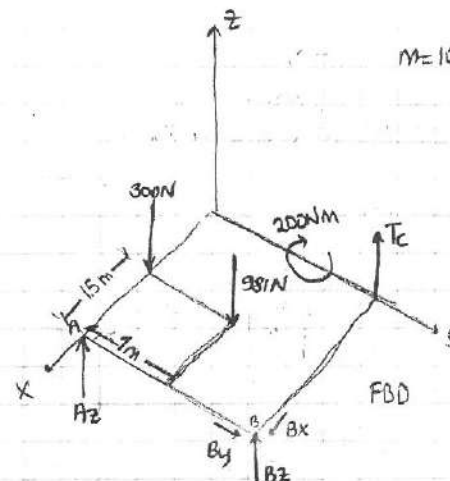
Four unknowns

Ex: The homogeneous plate shown in Fig.1 has a mass of 100kg and is subjected to a force and couple moment along its edges. If it is supported in the horizontal plane by a roller at A, a ball and socket joint at B, and a cord at C, determine the components of reaction at these supports.



Note! The moment of a force about an axis is equal to the product of the force magnitude and the perpendicular distance (moment arm) from the line of action of the force to the axis. Also, forces that are parallel to an axis or pass through it create no moment about the axis.

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$$M = 100 \text{ kg} \Rightarrow W = mg = (100 \text{ kg})(9.81 \text{ m/s}^2) = 981 \text{ N}$$

$$\begin{aligned} \sum F_x &= 0 & \sum M_x &= 0 \\ \sum F_y &= 0 & \sum M_y &= 0 \\ \sum F_z &= 0 & \sum M_z &= 0 \end{aligned}$$

$$\begin{aligned} \sum F_x &= 0 \Rightarrow B_x = 0 & (1) \\ \sum F_y &= 0 \Rightarrow B_y = 0 & (2) \\ \sum F_z &= 0 \Rightarrow A_z + T_c + B_z - 300 \text{ N} - 981 \text{ N} = 0 & (3) \end{aligned}$$

$$\begin{aligned} \sum M_x &= 0 \Rightarrow B_z(2 \text{ m}) + T_c(2 \text{ m}) - 981(1 \text{ m}) = 0 & (4) \\ \sum M_y &= 0 \Rightarrow -A_z(3 \text{ m}) - B_z(3 \text{ m}) + (300 \text{ N})(1.5 \text{ m}) + (981 \text{ N})(1.5 \text{ m}) - 200 \text{ Nm} = 0 & (5) \\ \sum M_z &= 0 \Rightarrow -B_x(2 \text{ m}) + B_y(3 \text{ m}) = 0 & (6) \end{aligned}$$

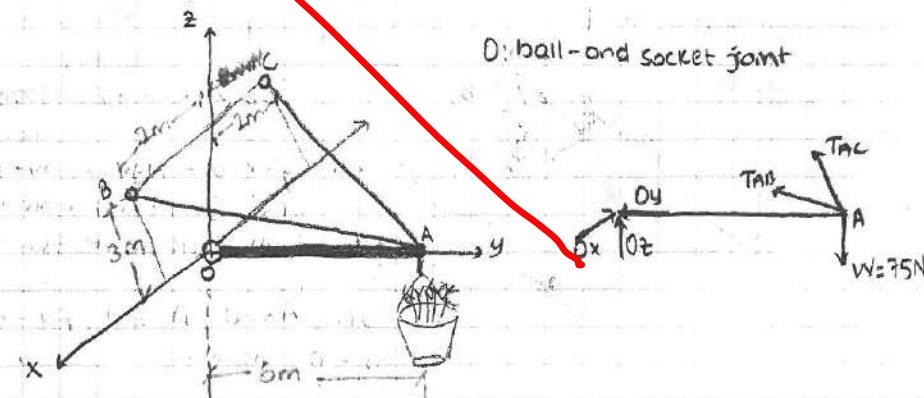
$$\sum M_x' = 0 \Rightarrow -A_z(2 \text{ m}) + (300 \text{ N})(2 \text{ m}) + (981 \text{ N})(1 \text{ m}) = 0 \quad A_z = 790 \text{ N}$$

$$\sum M_y' = 0 \Rightarrow T_c(3 \text{ m}) - (300 \text{ N})(1.5 \text{ m}) - (981 \text{ N})(1.5 \text{ m}) = 0 \quad T_c = 707 \text{ N}$$

$$\sum M_z' = 0 \Rightarrow A_z = 790 \text{ N}; B_z = -217 \text{ N}; T_c = 707 \text{ N}$$

Ex:

The boom is used to support the 75 N flowerpot in Fig.2. Determine the tension developed in wires AB and AC.



(7)

$\sum \vec{M}_O = 0$ (unit vector)

$\vec{T}_{AB} = T_{AB} \vec{\lambda}_{AB}$ $A(0, 6m, 0)$
 $\vec{T}_{AC} = T_{AC} \vec{\lambda}_{AC}$ $B(2m, 0, 3m)$
 $C(-2m, 0, 3m)$

$\vec{\lambda}_{AB} = \frac{2\vec{i} - 6\vec{j} + 3\vec{k}}{\sqrt{2^2 + (-6)^2 + 3^2}} = \frac{2}{7}\vec{i} - \frac{6}{7}\vec{j} + \frac{3}{7}\vec{k}$

$\vec{\lambda}_{AC} = \frac{-2\vec{i} - 6\vec{j} + 3\vec{k}}{\sqrt{(-2)^2 + (-6)^2 + 3^2}} = -\frac{2}{7}\vec{i} - \frac{6}{7}\vec{j} + \frac{3}{7}\vec{k}$

$\vec{T}_{AB} = T_{AB} \vec{\lambda}_{AB} = \frac{2}{7} T_{AB} \vec{i} - \frac{6}{7} T_{AB} \vec{j} + \frac{3}{7} T_{AB} \vec{k}$

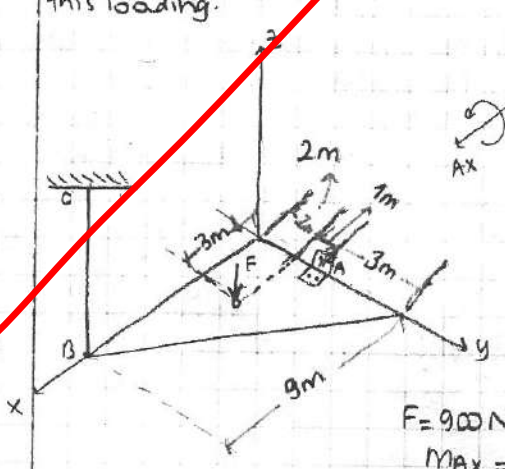
$\vec{T}_{AC} = T_{AC} \vec{\lambda}_{AC} = -\frac{2}{7} T_{AC} \vec{i} - \frac{6}{7} T_{AC} \vec{j} + \frac{3}{7} T_{AC} \vec{k}$

$\sum \vec{M}_O = 0 \Rightarrow (6\vec{j}) \times T_{AB} + (6\vec{j}) \times T_{AC} + (6\vec{j}) \times (\vec{W}) = 0 \quad (\vec{W} = -75\vec{k} \text{ N})$

$\vec{M} = \vec{r} \times \vec{F} \quad -\frac{12}{7} T_{AB} \vec{k} + \frac{18}{7} T_{AC} \vec{i} + \dots$

$T_{AB} = T_{AC} = 87.5 \text{ N}$

If the cable can be subjected to a maximum tension of 300N, determine the maximum force F which may be applied to the plate. Compute the x,y,z components of reaction at the hinge A for this loading.

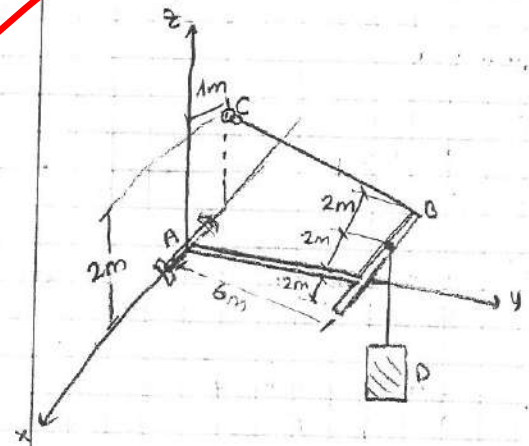


Free body diagram at A:
 $\sum F_x = 0 \Rightarrow A_x = 0$
 $\sum F_y = 0 \Rightarrow A_y = 0$
 $\sum F_z = 0 \Rightarrow A_z + 300N - F = 0$
 $\sum M_x = 0 \Rightarrow M_{Ax} - F(2m) + A_z(3m) = 0$
 $\sum M_y = 0 \Rightarrow F(3m) - 300N(9m) = 0 \Rightarrow F = 900N$
 $\sum M_z = 0 \Rightarrow M_{Az} - A_x(3m) = 0 \Rightarrow M_{Az} = 0$

$F = 900 \text{ N}; A_x = 0; A_y = 0; A_z = 600 \text{ N}$
 $M_{Ax} = 0; M_{Az} = 0$

(8) Ex:

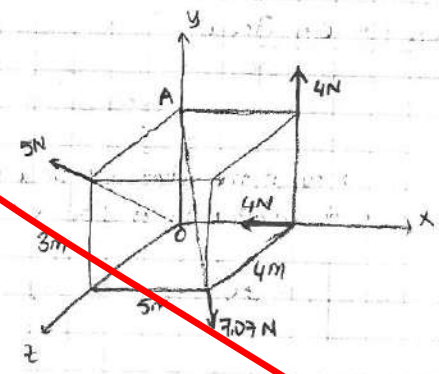
The member is supported by a pin at A and a cable BC. If the load at D is 900 N, determine the x,y,z components of reaction at the pin A and the tension in cable BC.



$T_{BC} = 1050 \text{ N}$
 $A_x = -450 \text{ N}$
 $A_y = 900 \text{ N}$
 $A_z = 0$
 $M_{Ay} = -600 \text{ Nm}$
 $M_{Az} = -900 \text{ Nm}$

Ex:

Four forces act on a rectangular parallelepiped as shown in Fig. 1. Reduce the system to a resultant force R applied at O and a resultant couple M.

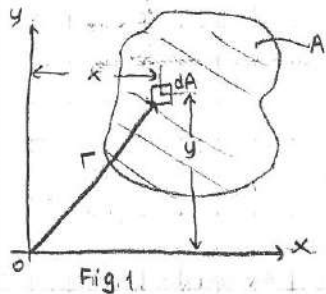


$R = 9 \text{ N}$
 $\alpha = 83^\circ 37'$
 $\beta = 63^\circ 37'$
 $\gamma = 27^\circ 16'$
 $M = 13 \text{ Nm}$
 $\alpha' = 22^\circ 33'$
 $\beta' = 90^\circ 00'$
 $\gamma' = 67^\circ 23'$

13.12.2016

Moment of Inertia: The moments of inertia of a differential area dA about the x and y axes are $dI_x = y^2 dA$ and $dI_y = x^2 dA$, respectively, Fig.1 For the entire area A the moments of inertia are determined by integration, i.e.,

$$I_x = \int_A y^2 dA \quad ; \quad I_y = \int_A x^2 dA$$

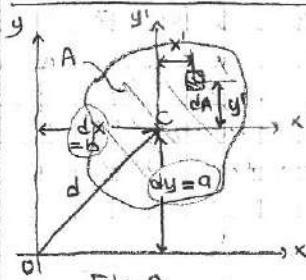


We can also formulate this quantity for dA about the "pole" O or z -axis, Fig.1

$$J_O = \int_A r^2 dA = \int_A (x^2 + y^2) dA = \int_A x^2 dA + \int_A y^2 dA$$

$$r^2 = x^2 + y^2 \quad \Rightarrow \quad J_O = I_x + I_y$$

Parallel-Axis Theorem for an Area: The parallel-axis theorem



can be used to find the moment of inertia of an area about any axis that is parallel to an axis passing through the centroid and about which the moment of inertia is known. (Fig.2)

$$I_x = \int_A y^2 dA = \int_A (y' + a)^2 dA = \int_A (y'^2 + 2ay' + a^2) dA$$

$$= \underbrace{\int_A y'^2 dA}_{I_{x'}} + \underbrace{2a \int_A y' dA}_0 + \underbrace{a^2 \int_A dA}_{a^2 A} \quad (y' = \bar{y} = 0)$$

$$I_x = I_{x'} + Aa^2$$

$$I_y = I_{y'} + Ab^2$$

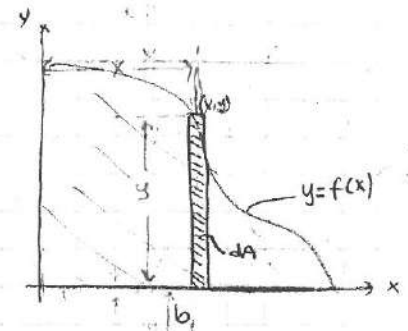
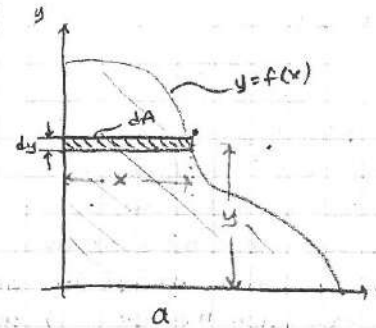
$$y = y' + dy = y' + a$$

$$x = x' + dx = x' + b$$

$$d^2 = a^2 + b^2$$

$$J_O = J_C + Ad^2$$

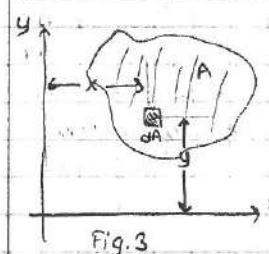
$$(I_x + I_y)$$



Moments of Inertia for Composite Areas: The moment of inertia of the entire area about the reference axis is determined by summing the results of its composite parts about this axis.

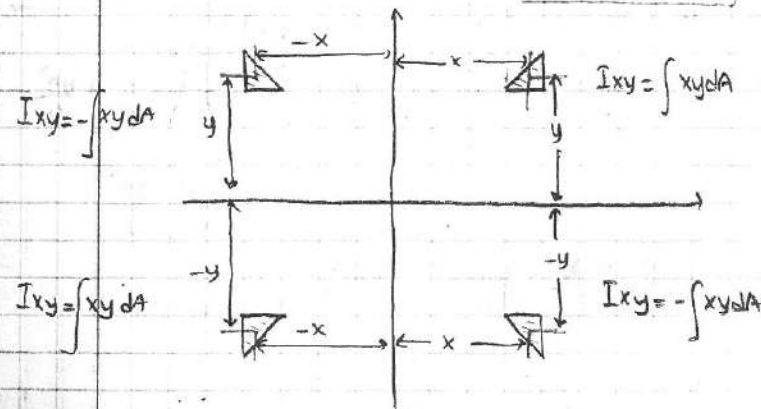
If a composite part has a "hole", its moment of inertia is found by "subtracting" the moment of inertia of the hole from the moment of inertia of the entire part including the hole.

Product of Inertia for an Area: The product of inertia is required



in order to determine the maximum and minimum moments of inertia of the area. The product of inertia of the area in Fig.3 with respect to the x and y axes is defined as

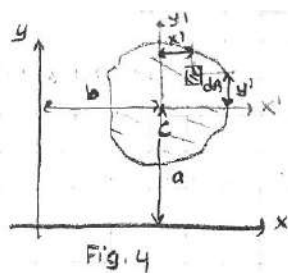
$$I_{xy} = \int_A xy dA$$



$$I_{xy} > 0$$

$$I_{xy} = 0$$

$$I_{xy} < 0$$

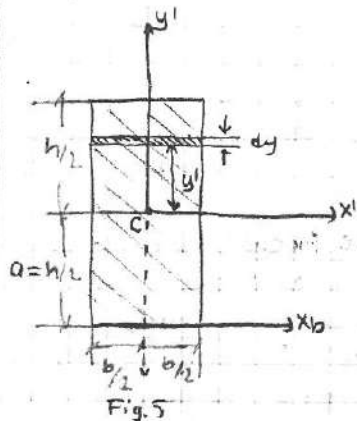


According to the parallel axis theorem:

$$I_{xy} = I_{x'y'} + A(ab)$$

Fig. 4

Ex:



Determine the moment of inertia for the rectangular area shown in Fig. 5 with respect to the centroidal x' axis, the axis x_b passing through the base of the rectangle, and the pole or z' axis perpendicular to the $x'-y'$ plane and passing through the centroid C .

$$dA = b dy' ; dA = h dx'$$

$$I_{x'} = \int_A y'^2 dA \Rightarrow I_{x'} = \int_{-h/2}^{h/2} (y')^2 b dy' = b \left(\frac{y'^3}{3} \right) \bigg|_{-h/2}^{h/2} = \frac{b}{3} \left(\frac{h^3}{8} + \frac{h^3}{8} \right) = \frac{bh^3}{12}$$

$$I_{y'} = \int_A x'^2 dA = \int_{-b/2}^{b/2} (x')^2 h dx' = h \left(\frac{x'^3}{3} \right) \bigg|_{-b/2}^{b/2} = \frac{h}{3} \left(\frac{b^3}{8} + \frac{b^3}{8} \right) = \frac{hb^3}{12}$$

using parallel axis theorem:

$$I_{x_b} = I_{x'} + A(a^2) = I_{x'} + A\left(\frac{h}{2}\right)^2 = \frac{bh^3}{12} + (hb) \left(\frac{h^2}{4}\right) = \frac{bh^3}{12} + \frac{bh^3}{4}$$

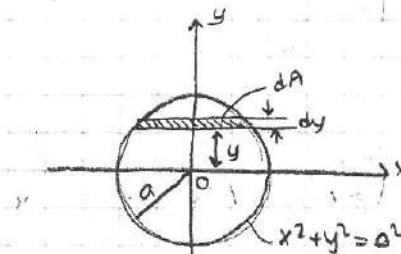
$$A = h \cdot b$$

$$= \frac{bh^3}{3}$$

$$J_c = I_{x'} + I_{y'} = \frac{bh^3}{12} + \frac{hb^3}{12} = \frac{bh}{12} (h^2 + b^2)$$

Ex:

Determine the moment of inertia with respect to the x -axis for the circular area shown in Fig. 6

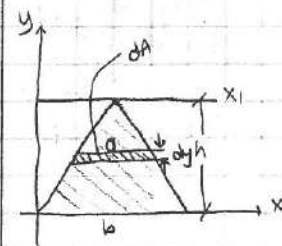


$$dA = (x - (-x)) dy = 2x dy$$

$$I_x = \int_A y^2 dA \Rightarrow I_x = \int_{-a}^a (y^2) 2x dy = \int_{-a}^a y^2 2\sqrt{a^2 - y^2} dy = \frac{\pi a^4}{4}$$

$$y = a \sin \theta \text{ (or } a \cos \theta = y)$$

15.12.2016

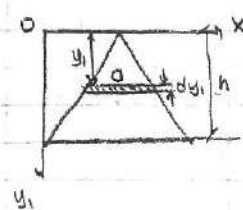


$I_{x_1} = ?$

$$dA = a dy \Rightarrow dA = \frac{b}{h} (h-y) dy$$

$$\frac{h-y}{h} = \frac{a}{b} \Rightarrow a = \frac{b}{h} (h-y)$$

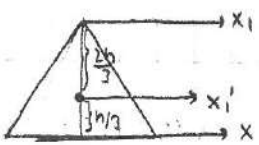
$$I_x = \int_A y^2 dA \Rightarrow I_x = \int_{y=0}^h (y^2) \frac{b}{h} (h-y) dy = \frac{bh^3}{12}$$



$$I_1 = \int_A y_1^2 dA = \int_0^h (y_1^2) \left(\frac{b}{h} y_1 dy_1 \right) = \frac{b}{h} \frac{y_1^4}{4} \bigg|_0^h = \frac{bh^3}{4}$$

$$\frac{y_1}{h} = \frac{a}{b} \Rightarrow a = \frac{b}{h} y_1$$

$$dA = \frac{b}{h} y_1 dy_1$$

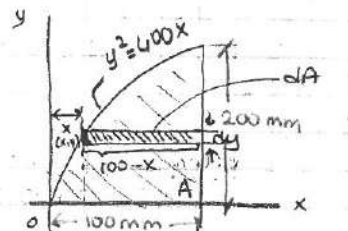


$$I_x = I_{x'} + A \left(\frac{h}{3} \right)^2 \Rightarrow I_x - \left(\frac{hb}{2} \right) \left(\frac{h^2}{9} \right) = \frac{bh^3}{12} - \frac{bh^3}{18} = \frac{bh^3}{36}$$

$$I_{x1} = I_{x'} + A \left(\frac{2h}{3} \right)^2 = \frac{bh^3}{36} + \left(\frac{hb}{2} \right) \left(\frac{4h^2}{9} \right) = \frac{9bh^3}{36} = \frac{bh^3}{4}$$

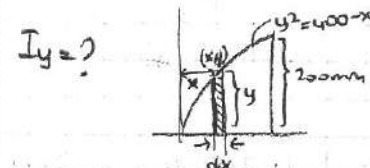
E:

Determine the moment of inertia for the shaded area shown in Fig.1 about the x-axis.



$$I_x = \int_A y^2 dA = \int_0^{200} (y^2) \left(100 - \frac{y^2}{400} \right) dy = 100 \frac{y^3}{3} - \frac{y^5}{5} \cdot \frac{1}{400} \Big|_0^{200} = 107.10^6 \text{ mm}^4$$

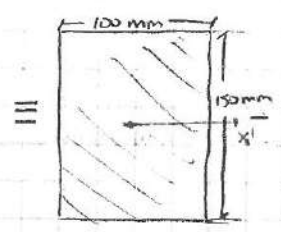
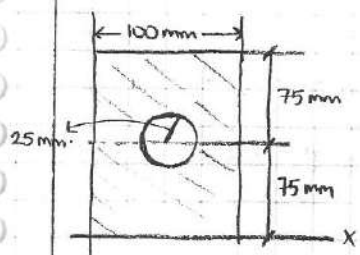
$$dA = (100 - x) dy = \left(100 - \frac{y^2}{400} \right) dy$$



$$I_y = \int_A x^2 dA = \int_0^{100} x^2 \sqrt{400x} dx = 20 \int_0^{100} x^{5/2} dx = 20 \cdot \left(\frac{2}{7} \right) x^{7/2} \Big|_0^{100} = 20 \cdot \left(\frac{2}{7} \right) 10^7 \text{ mm}^4$$

Ev:

Determine the moment of inertia of the area shown in Fig.1 about the x-axis.



$$I_x = I_{\text{rectangular}} - I_{\text{disc}}$$

b=100 mm
h=150 mm

$$I_{x \text{ rectangular}} = I_{x'} + A (75 \text{ mm})^2 = \frac{(100 \text{ mm})(150 \text{ mm})^3}{12} + (100 \text{ mm})(150 \text{ mm})(75 \text{ mm})^2$$

$$I_{x \text{ rec}} = 112.5 \times 10^6 \text{ mm}^4$$

$$I_{x \text{ disc}} = I_{x'} + A (75 \text{ mm})^2 = \frac{\pi (25 \text{ mm})^4}{4} + (\pi (25 \text{ mm})^2) (75 \text{ mm})^2 = 11.4 \times 10^6 \text{ mm}^4$$

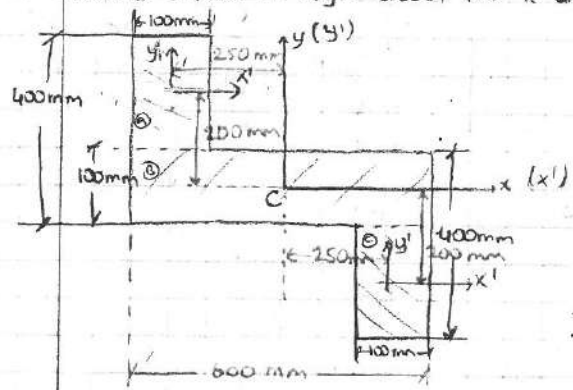
$$x = \frac{\pi a^4}{4}$$

a=25 mm

$$I_x = 112.5 \times 10^6 - 11.4 \times 10^6 = 101.1 \times 10^6 \text{ mm}^4$$

Ex:

Determine the moments of inertia for the cross-sectional area of the member shown in Fig.1 about the x and y centroidal axes.



rectangles A and C:

$$I_x = I_{x'} + A (200 \text{ mm})^2 = \frac{(100)(300)^3}{12} + (100)(300)(200)^2 = 1.425(10^9) \text{ mm}^4$$

$$I_y = I_{y'} + A (250 \text{ mm})^2 = \frac{(300)(100)^3}{12} + (100)(300)(250)^2 = 1.90(10^9) \text{ mm}^4$$

rectangle B:

$$I_x = I_{x'} + A (0)^2 = \frac{(600 \text{ mm})(100 \text{ mm})^3}{12} = 1.425(10^9) \text{ mm}^4$$

$$I_y = I_{y'} + A (0)^2 = \frac{(100 \text{ mm})(600 \text{ mm})^3}{12} = 1.90(10^9) \text{ mm}^4$$

$$I_x = I_{xA} + I_{xB} + I_{xC} = 2I_{xA} + I_{xB} = 2(1.425(10^9)) + 0.05(10^9) = 2.90(10^9) \text{ mm}^4$$

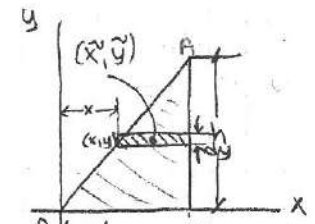
$$I_y = I_{yA} + I_{yB} + I_{yC} = 2I_{yA} + I_{yB} = 2(1.90(10^9)) + 1.90(10^9) = 5.60(10^9) \text{ mm}^4$$

$$I_{xy} = 0$$

product moment of inertia $\Rightarrow$ carpim eylesizlik momentleri bul!

Ex:

Determine the product of inertia I_{xy} for the triangle shown in Fig. 1.



$$I_{xy} = \int_A xy \, dA$$

$$dA = (b-x) dy = \left(b - \frac{by}{h}\right) dy$$

$$\bar{x} = x + \frac{b-x}{2} = \frac{b+x}{2}$$

$y = \frac{h}{b}x$
 $y = \frac{h}{b}x \Rightarrow x = \frac{b}{h}y$

$$y = \frac{h}{b}x$$

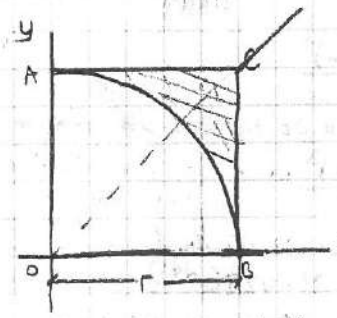
$$I_{xy} = \int_0^h \left(\frac{b+x}{2}\right)(y)(b-x) dy = \frac{1}{2} \int_0^h y(b^2 - x^2) dy = \frac{1}{2} \int_0^h y \left(b^2 - \frac{b^2 y^2}{h^2}\right) dy$$

$$= \frac{b^2}{2h^2} \int_0^h (yh^2 - y^3) dy = \left(\frac{b^2}{2h^2}\right) \left(\frac{yh^2}{2} - \frac{y^4}{4}\right) \Big|_0^h = \left(\frac{b^2}{2h^2}\right) \left(\frac{h^4}{2} - \frac{h^4}{4}\right)$$

$$= \frac{b^2 h^2}{8}$$

Ex:

Find the moment of inertia of the shaded area ACB in Fig. 1 about the x-axis.



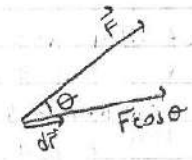
$$\left(I_x = \frac{r^4}{3} - \frac{\pi r^4}{16} \approx 0.137(r^4)\right)$$

76

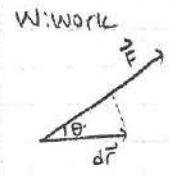
Virtual Work

20.12.2016

Work of a Force: A force does work when it undergoes a displacement in the direction of its line of action.



$$dW = F \, dr \cos \theta$$

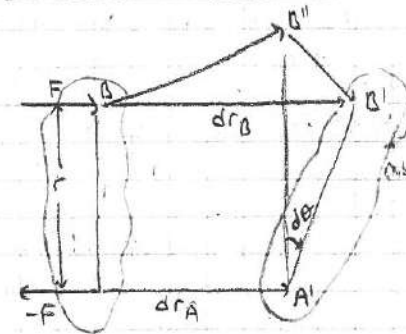


$$dW = \vec{F} \cdot d\vec{r}$$

$$= |\vec{F}| |d\vec{r}| \cos \theta$$

$$= F \, dr \cos \theta \quad (\text{scalar})$$

Work of a Couple moment: (A moment is vector quantity.)



The rotation of a couple moment also produces work.

$$dW = M \, d\theta$$

If $\vec{M}$ and $d\theta$ have the same sense the work is positive, however if they have opposite sense, the work will be negative.

A couple moment $\vec{M}$ having magnitude $F \cdot r = M$.

Virtual work: Consider now an imaginary or virtual movement of a body in static equilibrium, which indicates a displacement or rotation that is assumed and does not actually exist.

These movements are first-order differential quantities, and will be denoted by the symbols δr and $\delta \theta$ (delta r and theta).

The virtual work done by a force having a virtual displacement δr is

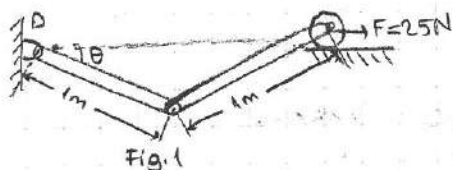
$$\delta W = F \cos \theta \, \delta r$$

Similarly, when a couple undergoes a virtual rotation $\delta\theta$ in the plane of the couple forces, the virtual work is

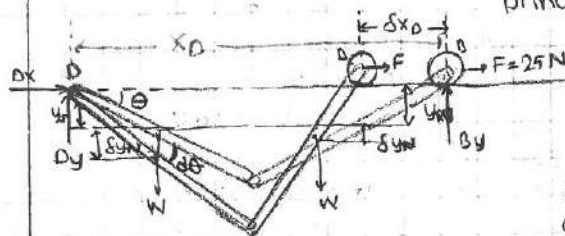
$$\delta W = M \delta\theta$$

Principle of virtual work: The principle of virtual work states that if a body is in equilibrium, then the algebraic sum of the virtual work done by all the forces and couple moments acting on the body, is zero for any displacement of the body.

$$\delta W = 0$$



Determine the angle θ for equilibrium of the two-member linkage shown in Fig. 1. Each member has a mass of 10 kg by using the principle of virtual work.



$$x_D = (1\text{m})\cos\theta + (1\text{m})\cos\theta = 2\cos\theta$$

$$y_D = \left(\frac{1}{2}\text{m}\right)\sin\theta = \frac{1}{2}\sin\theta$$

$$x_D = 2\cos\theta \Rightarrow \delta x_D = -2\sin\theta \delta\theta$$

$$y_D = \frac{1}{2}\sin\theta \Rightarrow \delta y_D = \frac{1}{2}\cos\theta \delta\theta$$

$$\delta W = 0 \Rightarrow +W(\delta y_D) + W(-\delta y_D) - F(-\delta x_D) = 0$$

$$2W(\delta y_D) + F(\delta x_D) = 0$$

$$2W\left(\frac{1}{2}\cos\theta \delta\theta\right) + F(-2\sin\theta \delta\theta) = 0$$

$$(W\cos\theta - 2F\sin\theta)\delta\theta = 0$$

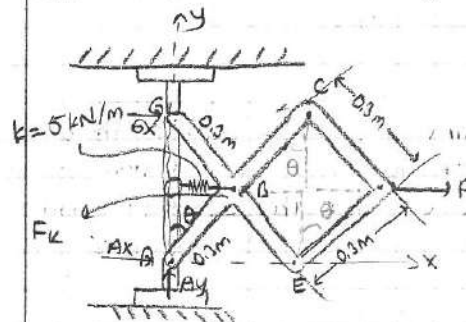
Arbitrary constant

$$\delta\theta \neq 0$$

$$W\cos\theta - 2F\sin\theta = 0 \Rightarrow \tan\theta = \frac{W}{2F}$$

EX:

Determine the required force P in Fig. 2, needed to maintain equilibrium of the scissors linkage when $\theta = 60^\circ$. The spring is unstretched when $\theta = 30^\circ$. Neglect the mass of the links.



$$F_k = k \delta x$$

$$F_k = (5\text{ kN/m})(0.3\sin\theta - 0.3\sin 30^\circ)$$

$$x_D = (0.3\text{m})\sin\theta$$

$$\Rightarrow \delta x_D = (0.3\text{m})\cos\theta \delta\theta$$

$$x_D = 3(0.3\text{m})\sin\theta$$

$$\Rightarrow \delta x_D = (0.9\text{m})\cos\theta \delta\theta$$

$$\delta W = 0 \quad -F_k(+\delta x_D) + P(+\delta x_D) = 0$$

$$-F_k(0.3\cos\theta \delta\theta) + P(0.9\cos\theta \delta\theta) = 0$$

$$\left[-((1.5)\sin\theta - (1.5)\sin 30^\circ)(0.3) + P(0.9) \right] \cos\theta \delta\theta = 0$$

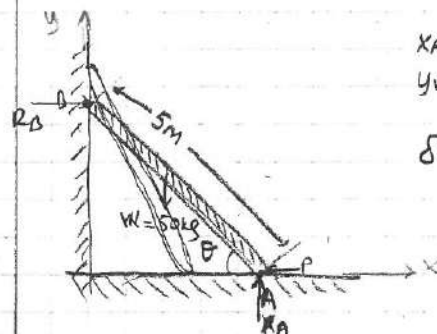
$$= 0 \quad \cos\theta \neq 0 \quad \delta\theta \neq 0$$

$$P = \frac{((1.5)\sin\theta - (1.5)\sin 30^\circ)(0.3)}{0.9} = \frac{1.5}{3}\sin\theta - \frac{1.5}{6}$$

$$\theta = 60^\circ \Rightarrow P = \left(\frac{1.5}{3}\sin 60^\circ - \frac{1.5}{6} \right) \text{ N}$$

EX:

Determine the magnitude of force P required to hold the 50 kg smooth rod in equilibrium at $\theta = 60^\circ$.



$$x_A = (5\text{m})\cos\theta \Rightarrow \delta x_A = -(5\text{m})\sin\theta \delta\theta$$

$$y_A = \left(\frac{5}{2}\text{m}\right)\sin\theta \Rightarrow \delta y_A = \left(\frac{5}{2}\text{m}\right)\cos\theta \delta\theta$$

$$\delta W = 0 \quad -W(+\delta y_A) + P(-\delta x_A) = 0$$

$$-W\left(\frac{5}{2}\cos\theta \delta\theta\right) + P(5\sin\theta \delta\theta) = 0$$

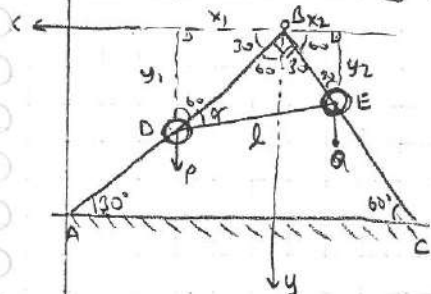
(79)

$$\left(-\frac{W}{2} \cos \theta + P \sin \theta\right) \delta \theta = 0$$

$$\delta \theta \neq 0$$

$$P \sin \theta = \frac{W}{2} \cos \theta \Rightarrow \frac{W}{2P} = \tan \theta \quad \theta = 60^\circ \Rightarrow P = \frac{W}{2 \tan 60^\circ}$$

Ex: Using the principle of virtual work, find the value of the angle α defining the configuration of equilibrium of the system shown in Fig. 3. The balls D and E can slide freely along the bars AC and BC but the string DE connecting them is inextensible.



$$\alpha \rightarrow \delta \alpha$$

$$\text{Active forces: } \vec{P} = P \vec{j}; \quad \vec{Q} = Q \vec{j}$$

$$\vec{r}_D = x_1 \vec{i} + y_1 \vec{j} = (l \cos \alpha) \cos 30^\circ \vec{i} + (l \cos \alpha) \sin 30^\circ \vec{j}$$

$$\vec{r}_E = -x_2 \vec{i} + y_2 \vec{j} = -(l \sin \alpha) \cos 60^\circ \vec{i} + (l \sin \alpha) \sin 60^\circ \vec{j}$$

$$\delta \vec{r}_D = -l \sin \alpha \cos 30^\circ \delta \alpha \vec{i} - l \sin \alpha \sin 30^\circ \delta \alpha \vec{j}$$

$$\delta \vec{r}_E = -l \cos \alpha \cos 60^\circ \delta \alpha \vec{i} + l \cos \alpha \sin 60^\circ \delta \alpha \vec{j}$$

$$\delta W_{\text{ext}} = 0$$

$$\delta \vec{W} = 0$$

$$\vec{P} \delta \vec{r}_D + \vec{Q} \delta \vec{r}_E = 0$$

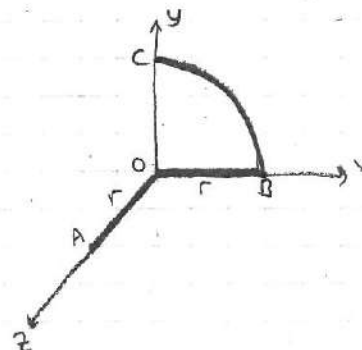
$$\tan \alpha = \sqrt{3} \frac{Q}{P}$$

(80)

Esra Sabancı
12059032

Quiz II

Locate the center of gravity of the slender homogeneous wire AOBDO

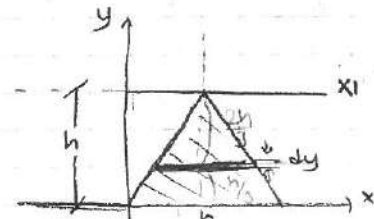


$$x_c = ?$$

$$y_c = ?$$

$$z_c = ?$$

Find the moment of inertia of the triangle in Fig 4 with respect to the axis x_1 .



$$dA = b dy$$

$$I_x = \int_A y^2 dA = \int_A y^2 b dy = b \frac{y^3}{3} \Big|_0^b$$